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Line Voltage-VF.CONCEPT

SFRS for Zefiro-Europe

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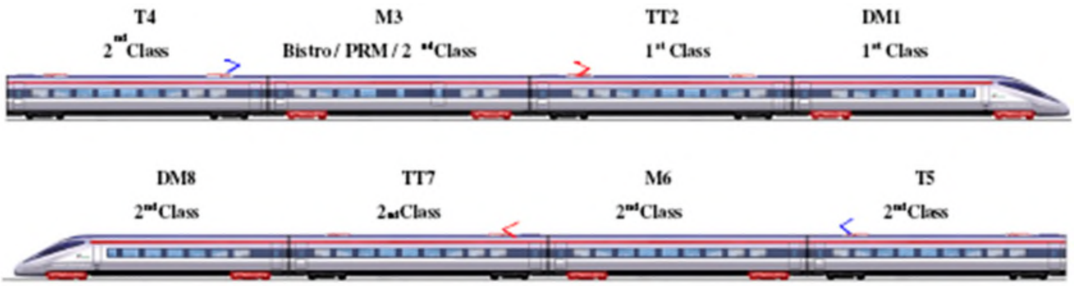
Content

1 Introduction

This document is a function specification for the control of the *Line Voltage system* for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- •DM = End driving motor car with driver's cab
- •TT = Intermediate trailer car with transformer
- •M = Intermediate driving motor car
- •T = Intermediate trailer car

1.2 Definitions and abbreviations

AC	Alternative Current
ADD	Automatic Dropping Device
ATP	Automatic Train Protection
CD	Circuit Diagram
DC	Direct Current
DCU	Drive Control Unit
DCUL	Drive Control Unit / Line
DI-signal	Digital Input signal
DO-signal	Digital Output signal
ETCS System	European Train Control System
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
FMECA	Failure Mode Effect and Criticality Analysis
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
HV	High Voltage
HW	Hardware
IDU	Interface Display Unit
LCB	Line Circuit Breaker
LCM	Line Converter Module
LIM	Line Interference Monitor
MCM	Motor Converter Module
MVB	Multifunction Vehicle Bus
N/A	Not Applicable
PCU	Propulsion Control Unit
PCU	Pantograph Control Unit
PGR	Passengers (Bombardier Division)
PPC	Propulsion & Controls (Bombardier Division)
PPU	Pantograph Pneumatic Unit
SFRS	System Function Requirement Specification
SIL	Safety Integrity Level

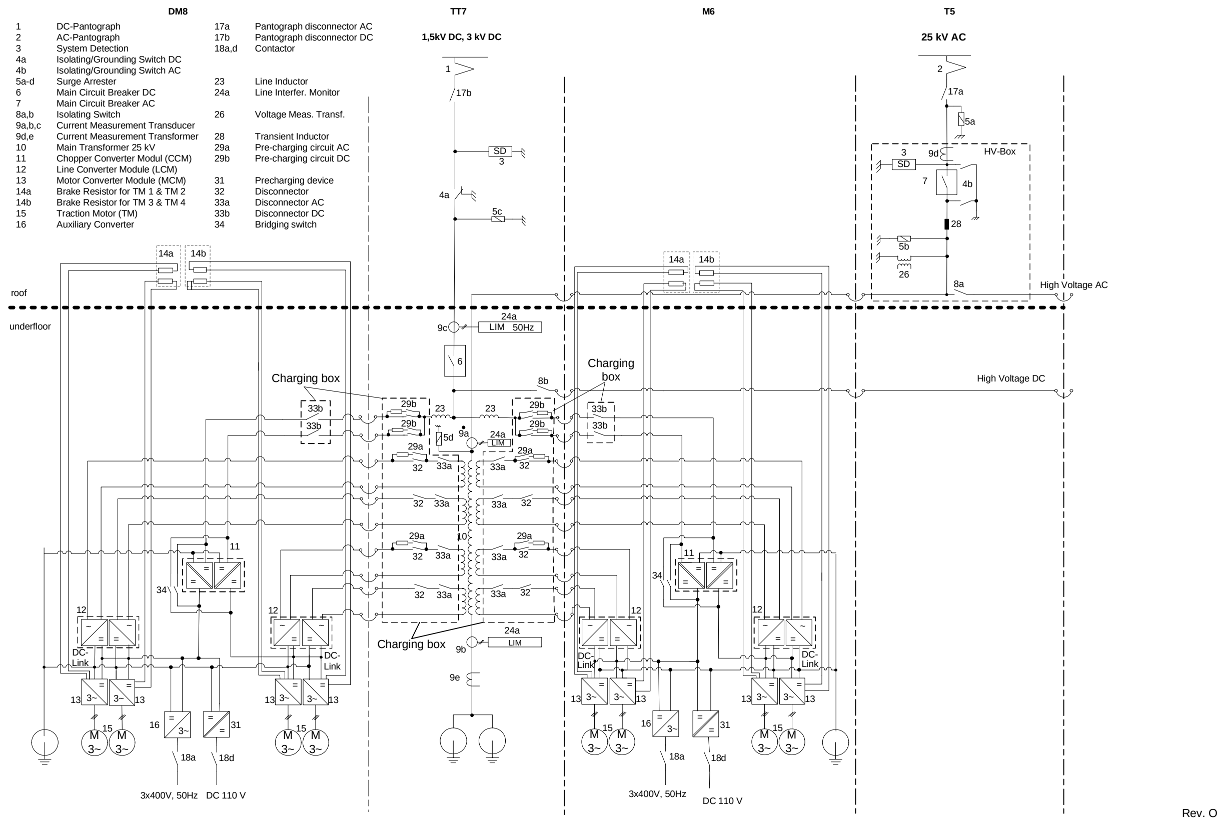
SIS	Safety instrumented system
SW	Software
TBD	To Be Defined
TCMS	Train Control and Management System
TRS	Technical Requirement Specification

All numbers in brackets refer to related numbers of system overview (see section 2.1).

2 System Overview

2.1 System Configuration

The architecture of the V300ZEFIRO energy distribution and traction/el. braking systems are designed with two Traction Units. Each traction unit consists of 4 coaches (half train), mirroring each other.



ID	2F_04.01.Zefiro-Europe.TRS.670
Description	The architecture of the power supply system shall support the following four different line voltage systems: AC 25 kV 50 Hz, AC 15 kV 16.7 Hz, 3 kV DC to 1.5 kV DC.
Safety level	N/A

The architecture of the power supply system supports the following four different line voltage systems: AC 25 kV 50 Hz, AC 15 kV 16.7 Hz, 3 kV DC to 1.5 kV DC.

The line voltage system is selected by the train driver (see chapter 3.3.5 "system selection"). To identify the country of operation, the driver has to select the country manually.
The High voltage system is divided into two sections: AC high voltage section and DC high voltage section. Both parts are built-up redundantly, that means they consist of two pantographs for each catenary system and identical subsystems. Both line voltage subsystems are connected via separate high voltage cables, thus each of the two pantographs can supply the whole trainset simultaneously. All pantographs and the system detection are able to withstand 25 kV AC.

3 Functional requirements

3.1 Country code selection

3.1.1 Function Design

3.1.1.1 Normal condition

Add description what the Country code is used for functions.

3.1.1.2 Failure condition

What kind of failure can it be with the country code selection

3.1.1.3 Failure supervision

How can failures be detected and what action must the system/driver do?

3.1.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.772
Description	It shall be possible to select the country which the train will be operated in from the IDU. Possible counties are: – Italy, Austria, Belgium, France, Germany, Netherlands, Spain and Switzerland. The country selection shall be used for country specific configurations such as: – selection of wide or narrow pantograph – levels of line voltage supervision – levels of line current harmonics.
Safety level	SIL 2

<table>
<tr><th>Nr</th><th>Description</th><th>Type</th><th>Derived to</th><th>User Interface</th><th>SIL</th></tr>
<tr><td>2F_04.01.Zefiro-Europe.CONCEPT.775</td><td>One of the following country-codes shall be selected by the driver on the IDU:
<table>
<tr><td>Code</td><td>Country of operation</td></tr>
<tr><td>1</td><td>Italy</td></tr>
<tr><td>2</td><td>Austria</td></tr>
<tr><td>3</td><td>Belgium</td></tr>
<tr><td>4</td><td>France</td></tr>
<tr><td>5</td><td>Germany</td></tr>
<tr><td>6</td><td>Netherlands</td></tr>
<tr><td>7</td><td>Spain</td></tr>
<tr><td>8</td><td>Switzerland</td></tr>
<tr><td></td><td></td></tr>
<tr><td>0</td><td>not defined</td></tr>
</table></td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td>IDU</td><td>SIL 2</td></tr>
<tr><td>2F_04.01.Zefiro-Europe.CONCEPT.965</td><td>The driver shall get a feedback on IDU from propulsion system that the system is prepared for operation in country or selected country-code. The driver has to confirm it on the IDU.</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td>IDU</td><td>SIL 2</td></tr>
<tr><td>2F_04.01.Zefiro-Europe.CONCEPT.1013</td><td>The selected country code shall be sent to all PCU's in the train.</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 2</td></tr>
<tr><td>2F_04.01.Zefiro-Europe.CONCEPT.797</td><td>In emergency operation mode (emegency operation mode trainline is energised) the latest selected/known country code shall be used and communicated to the PCU (LIM).</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 2</td></tr>
</table>

3.2 Pantograph Control

3.2.1 Pantograph - Lifting

3.2.1.1 Function Design

3.2.1.1.1 Normal Condition

Lifting the pantograph is a precondition for collecting energy from catenary system.
Lifting can be performed automatically or manually. In any case, the driver selects a line voltage system (4-position selector switch) and switches "raise pantograph 1" or "raise pantograph 2".
Country Code is needed to identify pantograph which is needed depending on country of operation.
The identified pantograph are lifted up on request.
TCMS sends to PCU by IP bus which type of pantograph has to be lifted. TCMS also commands 2 different DOs in order to define which type of pantograph has to be lifted.

3.2.1.1.2 Failure Condition

1) Pantograph doesn't lift up although an order was given.
2) Two pantographs of the selected type are lifting up. (unless for ice scraping)
3) Lifting of pantograph without order.
4) Wrong type of pantograph is lifting up.
5) Pantograph switch is in "up" position although related pantograph is lowered.

3.2.1.1.3 Failure Supervision

1) If the pantograph is not raised after 35 seconds after the raise order and the main reservoir pressure is higher than 750 kPa, the pantograph is considered faulty.
   If no air pressure is detected at the pantograph 5 minutes after the raise order and the main reservoir pressure is less than or equal to 750 kPa, the pantograph is considered faulty.
2) Both pantographs will be ordered down.
3) Pantograph will be lowered, considered faulty and blocked additionally an event will be set.
4) system detection detects invalid line voltage system or country code does not fit to raised pantograph. In both cases all pantographs will be lowered and a failure message will be displayed.
5) TCMS ignores pantograph-switch and pantograph remains in lowered position. IDU indicates that switch has to be turned to "down" position.

3.2.1.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.619
Description	The following sequence shall be implemented: * Driver selects the Country where the train is operated from the IDU. * Driver selects the line voltage system * The (not blocked or cut out) pantograph (one per 8 car train unit) for the chosen line voltage system shall be lifted when requested by the driver * Line voltage shall be detected and the chosen line voltage system verified If the detection and verification are successful, the closing of line circuit breakers shall be enabled. ELSE a diagnostic event shall be generated.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.735	With input from train driver (line voltage selector, see chapter 3.3.5 "System Selection") and depending on country of operation (country-code), TCMS shall select the needed type of pantograph (see chapter 3.2.5 "Pantograph selection").	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2

2F_04.01.Zefiro-Europe.CONCEPT.1099	If a pantograph control unit (PCU) is faulty an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1096	In case all pantograph control units (PCU) are faulty, TCMS shall communicate directly to pneumatic control unit (PPU) which pantograph shall be ordered to raise. The train speed shall be limited to 160 km/h.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1113	The backup raise order to the pneumatic control unit (PPU) unit shall be connected to a TCMS DO. When the DO is energized the EV1 (EV2) order shall be notified to the PCU via IP.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1114	The major failure indication from PCU shall be connected to a TCMS DI.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1168	If Major Fault from PCU has been detected in TCMS while the corresponding pantograph is raised, then "LCB emergency off" train line shall be immediately energized and the raising command shall be removed.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1174	Diagnosis of the contact: Major Fault (PCU) During the start up phase, after self test of PCU, when the life signal from PCU is being received, this test begins and TCMS shall manage it: – TCMS reads the contact: Major Failure in closed position (i.e no failure inside PCU); – TCMS sends for 6 s to PCU the command to open (just for test) the contact: Major Failure. This test is being carried out only if the pantograph is in lowered position. When the "Major Fault" contact test is on going, if the "Raise pantograph" switch is active, the raise order should be delayed to the end of the "Major Fault" contact test.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1175	After self test, when PCU receives from TCMS the demand to carry out the test of the contact: "Major Fault", it has to follow exactly the request from TCMS in order to sincronize this test. PCU opens its own "Major Fault" contact for 2 s for test purpose.	Derived Requirement	4S_04.01.02.-.Pantographs		
2F_04.01.Zefiro-Europe.CONCEPT.1185	While TCMS is carrying out the test of Major Fault contact (i.e. TCMS is sending to PCU the request to open that contact): – TCMS reads the the contact: Major Failure in open position (i.e. it is not stuck); – if the contact doesn't open during the test, at the end of this test an event shall be set (**Predisposition:** the affected pantograph shall be considered as "Faulty lowered")	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1237	When major failure test is performed the TCMS shall rise an event for engineering of 'major fault test ongoing'	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	
2F_04.01.Zefiro-Europe.CONCEPT.1236	The major failure contact test has shall be done also when master reset is ordered	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1097	If the pantograph control unit rises the variable 'not rised on command' an event shall be raised to the driver	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1121	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1156	When external power supply is active the pantograph shall not be allowed to raise.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.767
Description	If the "Pantograph 1" switch is turned to "up" position the front pantograph in the train shall be raised.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.980	The "up" position of the Pantograph 1 switch shall be connected to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.544	When driver activates "Raise pantograph 1" switch, the front pantograph of the needed pantograph type shall be raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_04.01.Zefiro-Europe.TRS.768
Description	If the "Pantograph 2" switch is turned to "up" position the rear pantograph in the train shall be raised.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.982	The "up" position of the Pantograph 2 switch shall be connected to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.736	When driver activates "Raise pantograph 2" switch, the rear pantograph of the needed pantograph type shall be raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_04.01.Zefiro-Europe.TRS.685
Description	For speeds greater than or equal to 300 km / h, the maximum number of pantographs raised shall be limited to two per train.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.766	When "Raise pantograph 2" switch is active and afterwards "Raise pantograph 1" switch will be activated, both pantographs of the selected pantograph type shall be raised. In multiple operation it is only allowed to raise one pantograph per train.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1058	In case front pantograph is raised and selector for rear pantograph is activated, rear pantograph shall not be allowed to raise.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1043	An event shall be set if front pantograph is raised and selector for rear pantograph is activated stating that “Ice scraping is only allowed if rear pantograph is raised first”.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.793	In multiple operation, when driver activates "Raise pantograph 1" switch, in both coupled trains the front pantograph of the needed pantograph type shall be raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_04.01.Zefiro-Europe.CONCEPT.1002	In a multiple coupled train: If pantograph 1 is selected to raise then the following actions shall be performed in priority order: 1) If both front pantographs are not ok then no pantograph shall be raised and an event shall be set for driver to change pantograph. 2) If front pantograph of first train is ok then front pantograph of first train shall be raised. 3) If front pantograph of second train is ok then front pantograph of second train shall be raised. 4) If front pantograph of first train is not ok and rear pantograph of first train is ok then rear pantograph of first train shall be raised. 5) If front pantograph of second train is not ok and rear pantograph of second train is ok then rear pantograph of second train shall be raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.794	In multiple operation, when driver activates "Raise pantograph 2" switch, in both coupled trains the rear pantograph of the needed pantograph type shall be raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1003	In a multiple coupled train: If pantograph 2 is selected to raise then the following actions shall be performed in priority order: 1) If both rear pantographs are not ok then no pantograph shall be raised and an event shall be set for driver to change pantograph. 2) If rear pantograph of first train is ok then rear pantograph of first train shall be raised. 3) If rear pantograph of second train is ok then rear pantograph of second train shall be raised. 4) If rear pantograph of first train is not ok and front pantograph of first train is ok then front pantograph of first train shall be raised. 5) If rear pantograph of second train is not ok and front pantograph of second train is ok then front pantograph of second train shall be raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1004	If in DC operation rear pantograph of first train and front pantograph of second train are raised there shall be a speed restriction of 300 kph (fine tuning to be done later) and an event for driver to change pantograph “Speed restriction due to pantograph fault. Change pantograph if high speed on DC line is required”	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.768	When an AC pantograph is not allowed to raise because there is already one up, this shall be indicated on IDU as an event.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.782	When any local pantograph-switch is in 'up' position at the cab activation (Note: not during parking exit procedure) an event to the driver shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.274	The related LCB (6,7) and system isolating switch (4a only in DC line voltage system) and AC earthing switch (4b only in AC line voltage system) must be in open position to lift the pantograph.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.563	When the pantograph is requested to raise, the pantograph disconnector (17) shall be closed first. When the feedback that the disconnector is closed is present, then the panto shall be ordered to raise.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1072	The pantograph disconnector shall close when ordered by TCMS DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

2F_04.01.Zefiro-Europe.CONCEPT.1159	Pantograph control unit (PCU) commands to lift the selected pantograph, only if there is congruence between information coming via IP bus and signals from digital input.	Derived Requirement	4S_04.01.02.-.Pantographs		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.281	In order the keep the pantograph up the pantograph main control valve shall be kept open by energizing the solenoid.	Derived Requirement	4S_04.01.02.-.Pantographs		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1173	In case the IP communication between Pantograph Control Unit (PCU) and TCMS has been missed, PCU is not allowed to block and the pantograph mustn't be lowered.	Derived Requirement	4S_04.01.02.-.Pantographs		
2F_04.01.Zefiro-Europe.CONCEPT.1115	The pantograph control unit (PCU) receives 2 different raise orders depending from the different type of pantograph. Each one shall be connected to a dedicated TCMS DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.542	If the air pressure to the pantograph is low (pressure switch) the auxliliary compressor feeding the selected pantograph shall be ordered to start when the order to raise the pantograph is given.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.543	If the air pressure to the pantograph is low (pressure switch) AND the main recevoir pressure is <6 bar; the order to raise the pantograph shall be set when: – pressure switch toggles to 'pressure available' OR – indipendently from the pressure switch after 6 minutes that the auxiliary compressor has been running continuosly.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1115	In case the pantograph rise order (at the first rising) is delivered based on the condition: ' after 6 minutes that the auxiliary compressor has been running continuosly' an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	
2F_04.01.Zefiro-Europe.CONCEPT.283	The raise order needs to be continuously high to raise and keep the pantograph in raised position. A raise order **must not be cancelled within the first 2 seconds** after activating it, else the pantograph structure might be harmed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1160	When DC line voltage system is selected and parking mode is activated, the second pantograph of same type has to be raised by TCMS without command from driver, but no change in LCB position. When AC line voltage system is selected and parking mode is activated, only the pantograph with the related LCB closed shall be kept raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.284	The pantograph status, with following priority order: – which pantos are "Cut out" – which pantos are "Faulty raised" – which pantos are "Faulty lowered" – which pantos are "Raised" – which pantos are "Lowered" shall be displayed on the IDU	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_04.01.Zefiro-Europe.CONCEPT.749	When a pantograph is raised, the information which pantograph is raised shall be transmitted to propusion system to enable activation of related high voltage equipment.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.737	When pantograph is raised, present line voltage system shall be detected by system detection (see chapter 3.3.5.2 "System selection").	Derived Requirement	4S_04.-.Power Supply		SIL 2

2F_04.01.Zefiro-Europe.CONCEPT.1107	For any pantograph configuration in multiple operation (AC or DC) a speed limitation has to be implemented. For now the limit will be equal to the max. test speed (final tuning will be done after first tests): • 396 kph in AC 25kV • 330 kph in AC 15kV • 330 kph in DC 3kV • 242 kph in DC 1.5kV	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.304
Description	To free the catenary line from ice it shall be possible to lift two pantographs per 8car train, however only one line circuit breaker is closed to supply the train with energy. In this case it is necessary to foresee a speed limitation.
Safety level	SIL 0

To free the catenary line from ice it will be possible to lift two pantographs per 8-car train, however only one line circuit breaker is closed to supply the train with energy.
In this case it is necessary to foresee a speed limitation.

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.783	For ice scraping it shall be allowed to turn both pantograph switches to "up" position when the train is operating in Italy (country-code = 1).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1238	When external temperature is <3°C **AND** train is running (V> 3km/h) **AND** the line selector is on 3kV a message 'risk of frozen catenary line' shall be raised to the driver after a time delay of 10 sec. the message shall be reset in the following cases: Ice scraping mode has been activated **OR** line selector set on AC mode **OR** external temperature is >=5°C for more than 10 sec	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.784	When both pantographs are raised for ice-scraping, the enable signal to close the LCB shall only be given to LCB which is related to the rear pantograph!	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.559	For ice scraping the pantograph closest to the active cab shall be used. NOTE! No ice scraping pantograph shall be used in a coupled train nor in backup mode.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.560	Ice scraping mode shall only be allowed in Italy.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

When a second pantograph is raised for ice scraping, it is in responsibility of the driver to limit the maximum speed at high speed lines to 200 km/h.

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.548	If two DC pantographs is raised for ice scraping, a warning message shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.484	The status of the ice scraping pantograph shall be displayed on the IDU in the same way as for a normal power collecting pantograph	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.486	If a second pantograph is raised (for ice scraping) then the LCB and the System Isolation Switch (only in DC) connected to the second pantograph raised shall be interlocked.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

2F_04.01.Zefiro-Europe.CONCEPT.748	If a second pantograph is raised (for ice scraping) then the LCB and the System Isolation Switch (only in DC) connected to the second pantograph raised shall be interlocked by software.	Derived Requirement	4S_04.-.Power Supply		SIL 0

3.2.2 Pantograph - Lowering

3.2.2.1 Function Design

3.2.2.1.1 Normal Condition

Lowering of pantograph, the Pantograph will stop collecting line current.
Pantograph will be lowered when line voltage system is being de-activated:

– by switch “Pantograph down” OR
– by drivers urgency (“Emergency push button”) push button OR
– by automatic device due to system change OR
– in case of critical failures (Pantograph control unit output “pantograph internal major failure” in case of Pantograph control unit fault or difference in pressure control)
– at cab deactivation (no parking mode active)

All line circuit breakers are ordered to open before lowering the pantograph except in emergency cases.

3.2.2.1.2 Failure Condition

1) Pantograph doesn't lower on order.
2) Pantograph is lowered unintentionally.

3.2.2.1.3 Failure Supervision

1) Manual lowering via panto-cock.
2) The faulty pantograph will be blocked.

3.2.2.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.266
Description	Pantograph shall be lowered when line voltage system shall be de-activated – by switching the selector switch to the down position OR – by drivers emergency push button OR – in case of failures. All line circuit breakers shall be open before lowering the pantograph except in emergency cases.
Safety level	SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.769
Description	If the "Pantograph 1" switch is turned to "down" position the front pantograph in the train shall be lowered.
Safety level	N/A

ID	2F_04.01.Zefiro-Europe.TRS.770
Description	If the "Pantograph 2" switch is turned to "down" position the rear pantograph in the train shall be lowered.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.984	The "down" position of the Pantograph 1 switch shall be connected to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

2F_04.01.Zefiro-Europe.CONCEPT.983	The "down" position of the Pantograph 2 switch shall be connected to TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.968	When driver activates "Lower pantograph 1" switch, the front pantograph of the needed pantograph type shall be lowered.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.969	When driver activates "Lower pantograph 2" switch, the rear pantograph of the needed pantograph type shall be lowered.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.561	When panto down order is given, the pantograph shall lower as soon as the LCB has opened or latest 3 seconds after the order was given.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1219	When a panto lowering order is given, a new rise command to the same pantograph shall be ignored untill the PCU delivers the information of panto lowered (Ipanpos = 0).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1220	In case a panto rise order is requested from the driver while the pantograph moving signal is set (during the lowering phase) an advisory should be displaied to driver asking to wait for the lowering phase to finish.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1225	When the pantograph is lowered (Ipanpos = 0), and the driver has tried to re-rise it during the lowering phase, an event shall be set to tell the driver that the panto is risable again.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.370	The pantograph shall not lower, except for specific emergency situations (stated in other requirements) if the LCB is closed or line current is detected.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.565	The pantograph disconnector shall be ordered to open after the pantograph has been lowered or at latest 10 seconds after the pantograph down order was given.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1073	The pantograph disconnector shall open when ordered by TCMS DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.77	In order to lower the pantograph the pantograph main control valve shall be switched over to drain the air pipes by removing the power to the solenoid.	Derived Requirement	4S_04.01.02.-.Pantographs		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.289	When lowering a DC pantograph the system isolation switch shall open when the pantograph is in down position.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.562	The pantograph shall be order to lower 500ms after an emergency stop button is pressed, regardless of the position of the LCB.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1098	The pantograph shall lower, using the fast lowering valve (panto fast down train line), 500ms after an emergency stop button is pressed, regardless of the position of the LCB.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.994	Any raised pantograph shall be lowered when requested from the PCU.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1008	If the cab is deactivated without entering the parking mode all pantographs shall be lowered.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1153	In case pantograph is lowered without order (signal from PCU) the LCB shall be opened by activation of "LCB emergency open" trainline, the pantograph order shall be removed and an alarm shall be set accordign to the diagnosys from the PCU.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

2F_04.01.Zefiro-Europe.CONCEPT.1165	In case Pantograph Control unit (PCU) detects incongruence between HW and SW 'raising signal' (= SW rise order is missing OR HW rise order is missing), the following actions shall be carried out. After 2 s from inchoerence detection: – **Predisposition:** LCB shall be ordered to open (TCMS via SW); – **Predisposition:** the pantograph shall be lowered and blocked – **Predisposition:** an event shall be set to the driver (Change pantograph required) – the related fault event shall be set to the maintainer (incoherence between HW and SW orders or vice-versa) After 16 s from inchoerence detection (if the pantograph has not been lowered): – **Predisposition:** TCMS shall energize "Panto Fast down " train line;	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1172	If the Pantograph Control Unit (of the panto in use) communication to the TCMS is NOT OK the following sequence shall be ordered: – LCB emergency open trainline shall be energized; – after 500 ms, Panto Fast down trainline shall be energized and and the raising command shall be removed;	Derived Requirement	4S_09.03.05.-.TCMS Software		

3.2.3 Automatic Dropping Device (ADD)

3.2.3.1 Function Design

3.2.3.1.1 Normal Condition

Lowering pantograph in case of collector head failure.

In order to have information about strips wearing there are two tubes inside each contact strip that are connected with pneumatic system, the tube at the highest level provides information about the wear of contact strips. The second tube at the lowest level activates the automatic lowering of pantograph (ADD).

When the ADD detects damages of contact strips it will drop the pantograph immediately.
In the event of collector head failure all pantographs of the train will be lowered immediately. After activation of ADD it must be possible to lift faultless pantographs.

3.2.3.1.2 Failure Condition

1) Strip is faulty but not detected by ADD.
2) Strip is not faulty but failure is detected by ADD.

3.2.3.1.3 Failure Supervision

1) No error-handling possible.
2) A faulty ADD-release will be handled as a normal ADD-release.

3.2.3.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.445
Description	The ADD shall detect damages of contact strips (rupture, abnormal wear) and insulated end horns.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.332	The ADD shall drop the pantograph immediately if any damage to the contact strips are detected.	Derived Requirement	4S_04.01.02.-.Pantographs		SIL 2

ID	2F_04.01.Zefiro-Europe.TRS.270
Description	In the event of collector head failure all pantographs of the train shall be lowered immediately. After activation of ADD it must be possible to lift faultless pantographs.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1061	When wear detection is activated, the Pantograph control unit (PCU) shall communicate to TCMS that a contact strip is worn.	Derived Requirement	4S_04.01.02.-.Pantographs		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1062	If wear detection is tripped an EVENT shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1063	If wear detection is tripped a counter shall be activated to measure the running distance with the worn pantograph in use.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1119	If worn pantograph is in use, TCMS shall send to PCU (pantograph control unit) the process data “Wear detection blocked”.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1064	The counter shall be allowed to reset to 0 by maintenance people when worn contact strip has been changed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.82	If ADD is tripped the status for the pantograph shown on the IDU shall be "faulty".	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.294	If ADD is tripped an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.295	If ADD is tripped the LCBs shall open immediately.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.747	If ADD is tripped the LCBs shall open immediately.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.371	If ADD is tripped the LCBs shall be requested to open immediately.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1178	If ADD is tripped and the LCB Emergency Off activation is not read in every line trip chain of whole composition an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1169	In case ADD has been detected, via the KPD relay, the raising command from TCMS has to be removed	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1170	Pantograph Control unit (PCU) resets the signal: "ADD tripped" after 30 s that the pantograph has been folded.	Derived Requirement	4S_04.01.02.-.Pantographs		
2F_04.01.Zefiro-Europe.CONCEPT.1171	The condition: "ADD tripped" is being reset in TCMS as soon as the signal (relay KPD) has gone to 0	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1067	In case of ADD tripped the pantograph has to be declared as faulty and the panto has to be blocked. The panto will be operational again only after a master reset.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1110	A block of pantograph due to ADD-trip shall be allowed to reset only by maintenance people.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

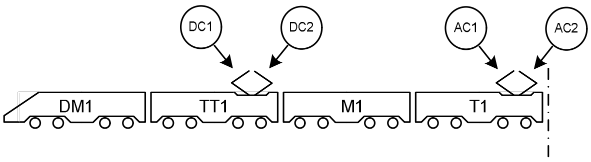
2F_04.01.Zefiro-Europe.CONCEPT.1068	If a faulty pantograph is blocked (due to ADD-trip), an EVENT shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.297	If ADD trips and train speed is > 0 km/h every pantograph in the whole train shall be lowered immediately via the "panto fast down" trainline.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1166	If "Panto Fast down" trainline is energized,TCMS sends this information to Pantograph Control Unit (PCU) via IP	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.296	If ADD trips and train speed is > 0 km/h every LCB shall requested to be opened and all pantographs in the whole train shall be lowered.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.372	The "fast panto down" trainline shall be set for 30 seconds when a ADD trip occurs.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1094	When pantograph raise order is given, activation of "fast panto down" trainline shall be delayed by n seconds to avoid that all pantographs will be lowered immediately.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.298	If the carbon strip is faulty the pneumatic control loop for the pantograph shall be drained.	Derived Requirement	4S_04.01.02.-.Pantographs		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.300	Insulating distance shall be reached immediately (worst case after 1s).	Derived Requirement	4S_04.01.02.-.Pantographs		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.301	After all pantographs have been dropped by ADD activating the "panto fast down" trainline, a non faulty pantograph shall be possible to select by the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1167	Deleted	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	
2F_04.01.Zefiro-Europe.CONCEPT.1176	To avoid false diagnosis, PCU has to acquire the status of EV3 (inside PPU) and consequently to mask the diagnostic signal "ADD tripped" if EV3 has been energized	Derived Requirement	4S_04.01.02.-.Pantographs		
2F_04.01.Zefiro-Europe.CONCEPT.1157	In case of failure of relays in ADD detection circuitry (unavailability for LCB opening and panto fast down train lines activation), the raising command for the pantograph shall be blocked and an EVENT shall be set. Note: this diagnosis has to be masked in case of "Panto Fast down" train line has been energized or any A-Key has been activated.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1158	Even though the pantograph was blocked due to failure recognised in ADD detection circuitry, in emergency operation mode the pantograph raising command shall be allowed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.2.4 Pantograph - Interlock

3.2.4.1 Function Design

3.2.4.1.1 Normal Condition

Faulty pantographs are blocked automatically. The driver can block a pantograph manually to prevent switching or if it is defect without automatic detection.



3.2.4.1.2 Failure Condition

1) Unidentified lifting of pantograph.

3.2.4.1.3 Failure Supervision

1) Pantograph will be lowered and blocked.

3.2.4.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.275
Description	The driver shall be able to manually block a pantograph.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.87	The driver shall be able to manually cut out a pantograph from selection	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_04.01.Zefiro-Europe.CONCEPT.303	If the pantograph is manually cut out an event shall be set as long as the pantograph is blocked.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.571	If a raised pantograph is manually cut out the LCB shall be requested to open and the pantograph shall lower.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.304	It shall not be possible to remove a manual cut out of the pantograph as long as any pantograph is raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.305	If a pantograph is blocked it shall always be considered down regardless of the pressure sensor indicates low or high pressure.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.306	If a pantograph is blocked the raise order shall be constant low.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_04.01.Zefiro-Europe.TRS.621
Description	A faulty pantograph shall automatically be blocked.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.207	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.309	In case the pantograph does not lower on command the diagnosys of the PCU shall rise an alarm to the driver and the pantograph shall be blocked.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.310	If the pantograph is in moving status (checking the pantograph position variable on IP) for t>20s an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.566	If the pantograph disconnector does not open when ordered the pantographs connected to the same LCB (DC1-DC2 or AC1-AC2) shall be permanently blocked and an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.567	If the pantograph disconnector does not close when ordered the pantograph shall be lowered, the pantographs connected to the same LCB (DC1-DC2 or AC1-AC2) shall be permanently blocked and an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

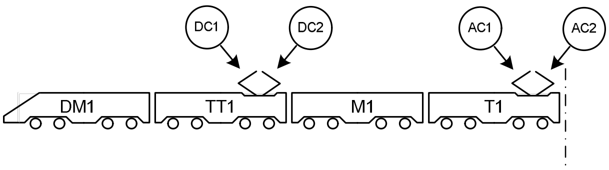
2F_04.01.Zefiro-Europe.CONCEPT.1103	If the pantograph disconnector opens without order ("closed" feedback disappears) while the pantograph is raised, the LCB shall be opened, the pantograph shall be lowered, the pantographs connected to the same LCB (DC1-DC2 or AC1-AC2) shall be permanently blocked and an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1074	Both the positions open and closed of the pantograph disconnector shall be connected to TCMS DI.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.2.5 Pantograph Selection

3.2.5.1 Function Design

3.2.5.1.1 Normal Condition

Pantographs of different type (AC and DC) and different width (1450mm, 1600mm and 1950mm) is used. Depending of the country the train operates in different width is required. Therefore, a country code to identify the current country of operations selected by the driver.



DC1 1450mm for DC-operation in Italy
DC2 1950mm for DC-operation in all countries but Italy
AC1 1450mm or 1600mm for AC-operation in Italy, France, Belgium, Switerland and Spain.
* 1450mm for operation in a corridor including Italy and Switerland.
* 1600mm for operation in a corridor including Italy and Spain.
* For Belgium and France both 1450mm and 1600mm can be used.
AC2 1950mm for AC-operation in Germany, Austria, Netherlands and Spain.

On the pure Italian train, only 2 different types of pantograph are installed: DC1 (1450mm) and AC1 (1450mm or 1600mm).

3.2.5.1.2 Failure Condition

1) Wrong pantograph type is ordered to lift up.
2) More than one pantograph gets raise order.

3.2.5.1.3 Failure Supervision

1) Driver supervises pantograph selection.
2) With exception of raising a second pantograph (ice scraping) all LCBs will open immediately and all pantographs will be lowered, if - in one 8 car train - more than one pantograph is raised.

3.2.5.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.286
Description	The 4 pantographs of one half-train (4 cars), which are used for AC voltage of 25 kV 50 Hz or 15 kV 16.7 Hz and DC voltage of 3 kV or 1,5 kV, shall be interlocked against each other. Switch over between the systems shall be performed automatically onboard the train set while the train is in motion.
Safety level	SIL 0

Only one pantograph on a full train set (8-cars) is selected for raising. The only situation when two pantograph selected is when one is used for ice scraping, see section 3.2.1

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.488	When voltage system AC 25kV or AC 15kV is selected and the train operates in Germany, Austria or Netherlands, the wider AC-pantograph (AC2) shall be selected.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 1
2F_04.01.Zefiro-Europe.CONCEPT.556	When voltage system AC 25kV or AC 15kV is selected and the train operates in Italy, Belgium or France, the narrow AC-pantograph (AC1, 1600mm or 1450mm depending on train configuration) shall be selected.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.786	When voltage system AC 25kV or AC 15kV is selected and the train operates in Spain, the narrow AC-pantograph (AC1, 1600mm) shall be selected.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.787	When voltage system AC 25kV or AC 15kV is selected and the train operates in Switzerland, the narrow AC-pantograph (AC1, 1450mm) shall be selected.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.557	When voltage system DC 3kV or DC 1.5kV is selected and the train operates in Belgium, France or Netherlands, the wider DC-pantograph (DC2) shall be selected.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.558	When voltage system DC 3kV or DC 1.5kV is selected and the train operates in Italy the narrower DC-pantograph (DC1) shall be selected.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.496	If the LCB connected to the selected pantograph is blocked, it shall be indicated on IDU and the driver will have to raise the other pantograph.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1095	When one of the following countries is selected all DC pantographs shall be blocked: Germany, Austria, Switzerland, Spain.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.325
Description	It shall be possible, through appropriate forcing procedures, to collect energy from any type of catenary by using the other AC pantograph of the trainset for AC lines or the other DC pantograph of the trainset for DC lines. This is limited for special purpose (like testing) under specific and supervised conditions.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.216	It shall be possible to control each pantograph individually by test parameters in the software. These parameters shall only be possible to be manipulated by connecting directly to the CCU with appropriate commissioning tools such as DCUTerm.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.2.6 Pantograph - Control of Static Contact Force

3.2.6.1 Function Design

3.2.6.1.1 Normal Condition

The static force of the raised pantograph will be controlled depending on speed, travel direction, position of pantograph and voltage system.

3.2.6.1.2 Failure Condition

1) Control system is not working correctly.
2) Ethernet link is missing.

3.2.6.1.3 Failure Supervision

1) Switching to other pantographs.
2) Switching to other pantographs.

3.2.6.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.454
Description	The pressure shall be a function of voltage system, speed, travel direction and position of pantograph
Safety level	N/A

The pressure is a function of voltage system, speed, travel direction and position of pantograph

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.97	The static contact force of the pantograph shall be regulated using the following parameters – Voltage system (AC or DC) – Train speed – Travel direction (Backwards and Forward) – Position of all raised pantographs in a train (counting from 1, closest to active cab Note: for a coupled consist the numbering continues) – applicability of ice-scraping mode	Derived Requirement	4S_04.01.02.-.Pantographs		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1111	TCMS shall provide following parameters to pantograph control unit (PCU): – Voltage system (AC or DC) – Train speed – Travel direction (Backwards and Forward) – Position of all raised pantographs in a train (counting from 1, closest to active cab Note: for a coupled consist the numbering continues) – applicability of ice-scraping mode	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.3 Supervision of Line Voltage and Line Current

3.3.1 Line Voltage Supervision

3.3.1.1 Function Design

3.3.1.1.1 Normal Condition

The supervision of the Line Voltage is done by the voltage sensors connected to the propulsion equipment. If the line voltage detection indicates a wrong line voltage system or a too high or too low Line Voltage, the active LCB will be opened respectively not closed.
Precondition: A country code is selected to identify the country of operation.

3.3.1.1.2 Failure Condition

1) No signal from system detection AC or DC (3).
2) Incorrect measurement (implausible voltage value) in system detection AC or DC (3).
3) No signal from AC voltage transformer (26)
4) Incorrect measurement (implausible voltage value) in AC voltage detection (26).

3.3.1.1.3 Failure Supervision

1) Driver has to change pantograph.
2) Driver has to change pantograph.
3) The error handling has to isolate the faulty half-train by opening of roof line disconnector 8a, refer to section 3.4.3.
4) The error handling has to isolate the faulty half-train by opening of roof line disconnector 8a, refer to section 3.4.3.

3.3.1.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.458
Description	Line voltage shall be supervised if it is within acceptable range to allow the line circuit breakers to close and to release full or reduced traction
Safety level	SIL 1

If "AC system detection" (inside HV-Box, T4 & T5 car) detects AC (50Hz or 16.7Hz) contact wire voltage, the voltage of AC-system will be measured in addition (15 kV or 25 kV).
If DC system detection (TT2 & TT7 car) detects DC contact wire voltage, the voltage of DC-system will be measured in addition (1.5 kV or 3 kV).

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_04.01.Zefiro-Europe.CONCEPT.222	The AC-LCB shall be allowed to close when the line voltage detection AC 50 Hz is ≥17.5 kV and ≤29.0 kV or line voltage detection AC 16.7 Hz is ≥11.0 kV and ≤18.0 kV.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1216	The LCB closing request has to be delivered to Propulsion only if: – the line voltage detection AC 50 Hz is ≥17.5 kV and ≤29.0 kV or line voltage detection AC 16.7 Hz is ≥11.0 kV and ≤18.0 kV – the line voltage detection DC 3kV is ≥1.975 kV and ≤3.9 kV (operation in Italy: ≤4.2 kV, operation in Belgium: ≤3.8 kV) or line voltage detection DC 1.5kV ≥1.0 kV and <1.975 kV	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1217	In case of requests to close the LCB with a measured line voltage not within the limits above an advisory shall be rised to the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	

2F_04.01.Zefiro-Europe.CONCEPT.313	The DC-LCB shall be allowed to close when the line voltage detection DC 3kV is ≥1.975 kV and ≤3.9 kV (operation in Italy: ≤4.2 kV, operation in Belgium: ≤3.8 kV) or line voltage detection DC 1.5kV ≥1.0 kV and <1.975 kV AND when system isolating switch (4a) is closed!	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.501	When DC line voltage system 3.0 kV is selected and the line voltage detection DC 3kV is ≥1.975 kV and ≤3.9 kV (operation in Italy: ≤4.2 kV, operation in Belgium: ≤3.8 kV) OR DC line voltage system 1.5 kV is selected and line voltage detection DC 1.5kV is ≥1.0 kV and <1.975 kV, the system isolating switch (4a) shall be closed.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.314	An event shall be set if the LCB cannot be closed due to the line voltage being out of the stated limits.	Derived Requirement	4S_04.-.Power Supply	EVENT	N/A

ID	2F_04.01.Zefiro-Europe.TRS.328
Description	When the maximum line voltage for the nominal 25kV system is exceeded; after 10 minutes operation at line voltage above 29kV or after 1 second at line voltage above 31 kV; the line circuit breaker shall be opened.
Safety level	SIL 2

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_04.01.Zefiro-Europe.CONCEPT.1082	If Line voltage AC 50 Hz is >32.0kV for ≥1 sec the pantograph shall be requested to lower and an event shall be set.	Derived Requirement	4S_04.-.Power Supply	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.315	If the AC LCB is closed and the Line voltage AC 50 Hz is >=31.1kV for ≥1 sec or >=29.1kV for ≥10 minutes the LCB shall be opened.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.578	If the AC LCB is closed and the Line voltage AC 50 Hz is >=31.1kV for ≥1 sec or >=29.1kV for ≥10 minutes an event shall be set.	Derived Requirement	4S_04.-.Power Supply	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.539	If the AC LCB is closed and the Line voltage AC 16.7 Hz is >18.5kV for ≥1 s the LCB shall be opened.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.579	If the AC LCB is closed and the Line voltage AC 16.7 Hz is >18.5kV for ≥1 s an event shall be set.	Derived Requirement	4S_04.-.Power Supply	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.319	The 10 minute clock for line voltage AC 50 Hz supervision shall have following behaviour: – line voltage >=29.1kV, clock will be counted-up – line voltage ≤29.0kV or LCB is open, clock will be resetted – - max. time is 10 minutes, min. value for clock is 0	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.316	If the DC LCB is closed and the Line voltage DC 3kV >4.1kV (operation in Italy: >4.2kV, operation in Belgium: >4.0kV) for ≥1 s or > 4.0kV for ≥5 minutes the LCB shall be opened.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.580	If the DC LCB is closed and the Line voltage DC 3kV >4.1kV (operation in Italy: >4.2kV, operation in Belgium: >4.0kV) for ≥1 s or > 4.0kV for ≥5 minutes an event shall be set.	Derived Requirement	4S_04.-.Power Supply	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.540	If the DC LCB is closed and the Line voltage DC 1.5kV >2.2kV for ≥1 sec the LCB shall be opened.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.581	If the DC LCB is closed and the Line voltage DC 1.5kV >2.2kV for ≥1 sec an event shall be set.	Derived Requirement	4S_04.-.Power Supply	EVENT	N/A

2F_04.01.Zefiro-Europe.CONCEPT.503	The 5 minute clock for line voltage DC 3 kV supervision shall have following behaviour: – line voltage >4.0kV, clock will be counted-up – line voltage ≤4.0kV, clock will be counted-down – max. time is 5 minutes, min. value for clock is 0	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.317	If the AC LCB is closed and the Line voltage AC 50Hz is <17,0 kV or Line voltage AC 16.7Hz is <10.5 kV for ≥1 sec the LCB shall be to open.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.582	If the AC LCB is closed and the Line voltage AC 50Hz is <17,0 kV or Line voltage AC 16.7Hz is <10.5 kV for ≥1 sec an event shall be set.	Derived Requirement	4S_04.-.Power Supply	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.318	If the DC LCB is closed and the Line voltage 3kV is <1.8 kV for ≥1 sec the LCB shall be opened.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.583	If the DC LCB is closed and the Line voltage 3kV is <1.8 kV for ≥1 sec the LCB shall be opened and an event shall be set.	Derived Requirement	4S_04.-.Power Supply	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.541	If the DC LCB is closed and the Line voltage 1.5kV is <0.85 kV for ≥1 sec the LCB shall be opened.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.584	If the DC LCB is closed and the Line voltage 1.5kV is <0.85 kV for ≥1 sec an event shall be set.	Derived Requirement	4S_04.-.Power Supply	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.321	The line voltage shall be displayed on IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.987	If line voltage is not detected within 5 seconds from when pantograph was indicated as raised an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.3.2 Line Current Supervision

3.3.2.1 Function Design

3.3.2.1.1 Normal Condition

AC line current measurement is done by line current transformer (9d) placed in AC HV-box on T4 and T5 car. It is connected to the LIM which is responsible for overcurrent detection.

DC line current measurement is done by DC line current sensor (9c).

At each main transformer a current transformer for line converter control is installed.

3.3.2.1.2 Failure Condition

1) No detection of line current when line current is expected.
2) No detection of transformer current when transformer current is expected.
3) Detection of line current when line current is not expected.

3.3.2.1.3 Failure Supervision

1) Event will be set and driver has to change pantograph.
2) Depending on the situation the transformer will be isolated, bypassed or event will be set. Handled by internal propulsion supervision.
3) Same action as LCB does not open on order

3.3.2.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.460
Description	At each main transformer there shall be a current transformer to measure the line current.
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.106	The current shall be measured by a return current transformer on the primary winding at each main transformer for the differential protection and the control of the LCMs. Note: Only applicable for AC operation	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.988	The transformer currents (input and return current) shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1129	If an differential overcurrent is detected, the LCB shall be opened and reclosing shall be blocked for 5 seconds.	Derived Requirement	4S_04.-.Power Supply		SIL 2

ID	2F_04.01.Zefiro-Europe.TRS.251
Description	Current measuring unit is used for line current measurement on high voltage side. The output signal shall be used for train protection (over currents, short circuits, differential currents) and for control purpose (line converter control).
Safety level	SIL 1

In AC line voltage system as well as in DC line voltage system there is a current measurement device installed in front of the LCB. The over current detection for AC is handled by the LIM and for DC by the LCB itself.

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1130	The line current shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.225	If an overcurrent is detected, the LCB shall be opened and reclosing shall be blocked for 5 seconds.	Derived Requirement	4S_04.-.Power Supply		SIL 1
2F_04.01.Zefiro-Europe.CONCEPT.361	First attempt of reclosing LCB: If overcurrent is detected within 5 seconds after first re-closure of the LCB, the LCB shall be opened again. The roof line disconnector related to the active pantograph shall be ordered to open. After reactivation of the line voltage system the same pantograph shall be used.	Derived Requirement	4S_04.-.Power Supply		SIL 1
2F_04.01.Zefiro-Europe.CONCEPT.362	second attempt of reclosing LCB: If overcurrent is detected within 5 seconds after second re-closure of the LCB, the LCB shall be opened again. The highvoltage system in the active train half shall be blocked and the other train half shall be used when next activation order is given. If overcurrent is not detected on second attempt there is no further action needed.	Derived Requirement	4S_04.-.Power Supply		SIL 1
2F_04.01.Zefiro-Europe.CONCEPT.363	If an overcurrent is detected an event shall be set.	Derived Requirement	4S_04.-.Power Supply	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.728	The maximum current drawn by the vehicle in line voltage system shall be limited according the value set by the driver on IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_04.01.Zefiro-Europe.TRS.773
Description	The maximum allowed power to be extracted from the catenary by the train (also in multiple) shall be possible to be set by the driver at the IDU.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1050	The driver shall be able to limit the train current by enabling a current limiter device on the IDU: a) in 25 kV AC line voltage systems by steps of 20 A (in range of 20 to 560 A) b) in DC 3 kV line voltage systems by steps of 100 A (in range of 100 to 3000 A) c) in DC 1.5 kV line voltage systems by steps of 100 A (in range of 100 to 3200 A) In case of 15 kV line voltage system: 700 A In case of double traction the limits are doubled.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1051	Current shall be limited according to given values from TCMS.	Derived Requirement	4S_04.-.Power Supply		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1152	TCMS shall set maximum level of current limitation after powered on: a) in 25 kV AC line voltage systems: 560 A b) in DC 3 kV line voltage system: 3000 A c) in DC 1.5 kV line voltage system: 3200 A In case of 15 kV line voltage system: 700 A In case of double traction the limits are doubled. Every time the catenary change, the max value for that catenary must be setted.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.3.3 Line Interference Monitor

3.3.3.1 Function Design

3.3.3.1.1 Normal Condition

Line interference monitor supervises interference currents (e.g. 50Hz supervision in Italian DC line), see details in TRD appendix Q.

3.3.3.1.2 Failure Condition

1) Failure to detect line voltage.
2) Failure to measure overcurrent.
3) Other failure of LIM.

3.3.3.1.3 Failure Supervision

1) Changeover to other HV system. Supervision by PCU.
2) Changeover to other HV system. Supervision by PCU.
3) LIM is bypasses, supervision by PCU.

3.3.3.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.309
Description	One line interference monitor per half-train shall be installed. (e.g. 50 Hz supervision in Italian DC line.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.112	At least one line interference monitor per half-train shall be installed for the supervision of the harmonic content for the DC high voltage line and for the AC high voltage line. See details in TRD appendix Q.	Derived Requirement	4S_04.-.Power Supply		N/A

ID	2F_04.01.Zefiro-Europe.TRS.691
Description	The LIM shall realize the following functions: – Line current measurement – Return current measurement – Line voltage measurement – Supervision of Transformer oil temperatures – Self test
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.231	The LIM shall realize the following functions: – Line current measurement AC – Return current measurement – Line voltage measurement – Self test	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.989	The LIM Self test shall be requested when: The Pantographs are lowered and the LCBs are open and any of the following conditions are fulfilled: – Train has started up (after closing of BC1) AND more than **15 hours** have passed since last LIM self test. – The self test request is set from the LIM and the train is at standstill and no cab is active. – The self test timeout is set from LIM. In this case the LCB shall be ordered to open and the pantograph shall be lowered in the same 8-car train. – **The cab is activted to exit of parking mode and more than 40 hours have passed since last LIM selt test. In this case the LCB shall be ordered to open and the pantograph shall be lowered in the train. An event shall be set to inform the driver** The self test request will be set **40hours** after last successful test. The self test timeout will be set after 50 hours and the LIM will start the test independent of TCMS. Info: It takes approximatley 45s for the LIM to start after power on. It takes approximately 30s for the LIM to perform the test but in case of retry from LIM a time out should be min 75s.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

2F_04.01.Zefiro-Europe.CONCEPT.1131	As long as the LIM self test in ongoing the LCB shall not be enabled to close and the pantograph shall not be raised. If the driver request to raise the pantograph while the LIM self test is ongoing an event shall be set "LIM self test ongoing" to inform the driver that the pantograph will be raised when the test is finished. The event shall be reset when LIM self test is finished or the driver request to raise the pantograph is removed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1132	An event shall be set "LIM self test finished" when the LIM self test is finished to inform the driver that it is possible to raise the pantograph and to close the LCB.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1087	In case of LIM self test error, LIM will be bypassed and pantograph isolated (100% traction effort remaining).	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1088	In case LIM self test will not be performed although it is requested, LIM will be bypassed and pantograph isolated (100% traction effort remaining).	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.756	If an self test error is indicated, an EVENT shall be set.	Derived Requirement	4S_04.-.Power Supply	EVENT	N/A

ID	2F_04.01.Zefiro-Europe.TRS.692
Description	The line circuit breaker shall open if – limits for line current exceeded (limits depending on country of operation) – Main transformer differential current exceeded – Short circuit detected – maximum line voltage exceeded
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.234	The line circuit breaker shall open if – limits for line overcurrents in AC exceeded – Main transformer differential current exceeded – Short circuit detected – maximum line voltage exceeded	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.757	An EVENT shall be set if limits for line overcurrent exceeded.	Derived Requirement	4S_04.-.Power Supply	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.758	An EVENT shall be set when main transformer differential current exceeded.	Derived Requirement	4S_04.-.Power Supply	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.759	An EVENT shall be set when short circuit detected.	Derived Requirement	4S_04.-.Power Supply	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.760	An EVENT shall be set when maximum line voltage exceeded.	Derived Requirement	4S_04.-.Power Supply	EVENT	N/A

3.3.4 LIM 50Hz protection bypass

3.3.4.1 Function Design

In case of detection of 50Hz current harmonic when the train is running on a DC line, the LIM protection system opens the line circuit breaker to avoid potential risks.
Most probably this is not due to a failure of the train propulsion system but to some incoming noise directly from the power line.

The Italian regulation states that is possible to bypass this protection allowing the train to move to a different segment of line where this problem is not present.
This is allowed only when the train is running on a DC powered line.
A sealed switch is installed in the driver's cab to let the driver perform this action. The activation of this switch is managed by TCMS safe layer that communicates directly to the LIM system to bypass this protection.

3.3.4.1.1 Normal Condition

In normal condition the switch is sealed and the 50Hz supervision functionality is not bypassed.

3.3.4.1.2 Failure Condition

1) The switch is turned but no bypass has been enabled.
2) The switch is not turned and the bypass has been enabled.

3.3.4.1.3 Failure Supervision

1) By the driver, if the protection occurs with the switch turned and the LCB cannot be closed.
2) By the driver, if the IDU indication is on without switch turned.

3.3.4.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.792
Description	A sealed switch to bypass the LIM 50Hz harmonic control (on DC lines only) shall be installed in the driver's cab.
Safety level	

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1146	A sealed switch to bypass the LIM 50Hz harmonic control (on DC lines only) shall be installed in the driver's cab.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1147	The LIM bypass switch shall be connected to safe and redundant Digital Inputs.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1148	When the sealed switch is turned TCMS shall send the "Command: Disable LIM 50Hz supervision" to Propulsion system via MVB.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1149	When the LIM bypass is activated an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1150	When the LIM bypass is activated an indication shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1151	The 50Hz supervision of LIM (on DC line only) shall be bypassed when commanded by TCMS. This has to be done also in multiple composition. Once the by pass is activated in one cab this by pass has to be done on both vehcles of the multiple composition.	Derived Requirement	4S_04.-.Power Supply		SIL 2

3.3.5 Neutral Section Passing

3.3.5.1 Function Design

3.3.5.1.1 Normal Condition

Neutral sections are powerless parts of the catenary system. There are two different types of neutral sections which are defined according to EN50388:

a) AC phase separation sections, which separate between two different phases of AC line voltage system.
   In Italy, crossing AC phase separation sections will be performed automatically by indications from ATP.

b) system separation sections, which separate between different line voltage systems (AC 25kV, AC 15kV, DC 3kV, DC 1.5kV)
In Italy, crossing system separation sections will be initiated by ATP but driver has to select line voltage manually.

3.3.5.1.2 Failure Condition

1) AC phase separation section is not detected (ATP is faulty/isolated).
2) System separation section is not detected (ATP is faulty/isolated).
3) AC phase separation section detected but LCBs do not open.
4) System separation section detected but LCBs do not open.
5) System separation section detected but pantograph does not lower

3.3.5.1.3 Failure Supervision

1) Driver has to open LCB manually (Open LCB switch) before entering neutral section.
2) Driver has to open LCB manually (Open LCB switch) and lower the pantograph before entering neutral section.
3) Driver has to open LCB manually (Open LCB switch) before entering neutral section. When LCB is faulty, panto will lower immediately.
4) Driver has to open LCB manually (Open LCB switch) before entering neutral section.
When LCB is faulty, panto will lower immediately.
5) Driver has to lower pantograph manually (Pantograph switch to “down position”) after LCB opened but before entering neutral section.

3.3.5.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.463
Description	The AC phase separation sections and separation section sequences shall be activated by External sensors e.g. ATP or track side magnets.
Safety level	SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.464
Description	When the AC phase separation section sequence is activated, the Line Converters shall be ramped down and the line circuit breakers shall be opened for the time of passage of the AC phase separation sections.
Safety level	SIL 0

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_04.01.Zefiro-Europe.CONCEPT.515	AC phase separation section sequence shall be activated by external sensors (e.g. ATP).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1112	System separation section sequence shall be activated by external sensors (e.g. ATP).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1122	In italy, the sequence of crossing AC phase separation sections shall start when external sensors orders “traction block” and “LCB opening”.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1123	In Italy, the sequence of crossing system separation sections shall start when external sensors orders “traction block”, “LCB opening” and “pantograph down”.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.521	For country options, the sequence of crossing AC phase separation sections or system separation sections shall start when the train has travelled the distance between the sensor in the leading unit and the raised pantograph.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_04.01.Zefiro-Europe.CONCEPT.510	The power reference shall be ramped down to 0 before requesting the LCB to open prior to enter AC phase separation section or system separation section.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.441	The LCB shall be requested to be opened before entering the AC phase separation section or system separation section.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.442	When the LCB has opened for AC phase separation section or system separation section it shall stay open for at least 4 seconds.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.512	AC phase separation section or system separation section sequence shall be reset if the train is standing still.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.513	The AC phase separation section or system separation section sequence shall be reset on the return of line voltage (positive edge)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1124	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1125	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.344	If LCB was opened on order from external sensors (ATP) before entering the AC phase separation section, the LCB shall be requested to be closed again after passing the AC phase separation section, automatically when ordered by external sensors only if any command to open hasn't been given from driver or internal protection of HV equipment and propulsion system, and system detection has confirmed selected line voltage system (see chapter 3.4.1).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.776	If LCB was opened on order from external sensors (ATP) before entering the system separation section, the LCB shall be requested to be closed again after passing the system separation section, automatically when ordered by external sensors only if any command to open hasn't been given from driver or internal protection of HV equipment and propulsion system, and system detection has confirmed selected line voltage system (see chapter 3.4.1).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.443	After the LCB has been closed automatically when ordered by external sensors (ATP), MCM power reference shall be ramped up.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_04.01.Zefiro-Europe.TRS.686
Description	In separation sections (neutral sections where supply voltage changes) the panthograph shall be lowered.
Safety level	N/A

Passing a system separation section will be done in the same way as a AC phase separation section with the extra function to swap pantograph before reconnection on a different line voltage system.

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.505	If system separation section is activated, from external sensors (ATP), LCB shall be ordered to open and pantograph shall be ordered to lower.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

The only action which is needed from driver during crossing of system separation section is the manual selection of the upcoming line voltage system (line voltage selector switch).

ATP will indicate to the driver which line voltage system will follow after the separation section.

ID	2F_04.01.Zefiro-Europe.TRS.708
Description	The AC phase separation or separation section shall be activated via External sensors (e.g. ATP). The External sensors also gives additional information to lower pantograph after the Line circuit breaker has been opened.
Safety level	SIL 0

The Italian ATP system reports: Information about the line voltage system on the other side of the neutral section.

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.458	If external sensors (ATP) indicate a AC phase separation section requires lowering of pantographs, all pantographs shall be lowered before entering the AC phase separation section. (Note this is not the same as a system separation section)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.573	If external sensors (ATP) indicate that the voltage system after passing the AC phase / system separation section is different than the voltage system before the AC phase / system separation section, the passage shall be done according to requirement for system separation section.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.335
Description	Where AC phase separation sections require pantographs to be lowered and subsequently raised, these additional actions are permitted to be automatically initiated by the ATP.
Safety level	SIL 0

External sensors (ATP) might in some situation be able to not only order pantograph down but also pantograph up.

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.523	If external sensors (ATP) require the pantograph to be raised, the selected pantograph shall raise. (This shall work in direct parallel to the "raise pantograph switch")	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1089	If the pantographs have been lowered on order from external sensor before entering a AC phase / system separation section and driver has not moved panto-switch to "down" position, raise-order for pantograph shall be blocked as long as the train has not passed neutral section.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1126	Pantograph shall only be allowed to raise, when TCMS has confirmed that selected line voltage system is in line with system which is indicated by ATP.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1127	It shall be indicated on IDU when selection of line voltage is not in line with ATP notification.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.462	If the pantographs have been lowered on order from external sensor before entering a AC phase / system separation section and driver has not moved panto-switch to "down" position, raise-order for pantograph after crossing the neutral section shall be set automatically by external sensors depending on selected line voltage system.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.687
Description	When multiple pantographs are lifted and spacing allowed in coming neutral section is exceeded, the pantographs shall be lowered before entering the neutral section.
Safety level	SIL 0

If two or more pantographs are lifted, a neutral section with a length about the same as the distance between two pantographs can be short circuit by the carbon strip connecting the neutral catenary with the live catenary.

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.577	At multiple operation in Italy: If the ATP has requested the pantograph to lower in the first train unit then the pantograph on the second train unit shall be lowered before entering a neutral section, i.e. when the train has travelled the distance between the ATP request and the second pantograph.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.518	In multiple operation, pantographs shall raise when external sensors order “pantograph raising”.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.3.6 System Selection

3.3.6.1 Function Design

3.3.6.1.1 Normal Condition

The driver selects the line voltage system (AC 25kV, AC 15kV, DC 3kV or DC1.5kV) by using 4-position switch "Line voltage selector".
When line voltage system changes from one to another of these four voltage systems a system selection has to be performed. To identify exactly the country of operation, the country code needs to be selected.
The system detection units have to detect, to which line voltage system the train is connected.

3.3.6.1.2 Failure Condition

1) System detected by AC system detection unit and LIM differ
2) Faulty System detected by AC system detection unit
3) System detected by DC system detection unit and LIM differ
4) Faulty System detected by DC system detection unit
5) Plausibility fault DC and 3kV from LIM

3.3.6.1.3 Failure Supervision

1) Faulty system detection. Lower and isolate AC pantograph. Driver has to raise other AC pantograph,
2) Faulty system detection. Lower and isolate AC pantograph. Driver has to raise other AC pantograph,
3) Faulty system detection. Lower and isolate DC pantograph. Driver has to raise other DC pantograph,
4) Faulty system detection. Lower and isolate DC pantograph. Driver has to raise other DC pantograph,
5) Faulty signals from LIM. Lower and isolate DC pantograph. Driver has to raise other DC pantograph,

3.3.6.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.771
Description	The following power supply system shall be possible to be selected by the driver at driver's desk: AC 25 kV 50 Hz, AC 15 kV 16.7 Hz, 3 kV DC, 1.5 kV DC.
Safety level	

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.492	The driver shall be able to select the voltage system (15kV AC, 25kV AC, 3kV DC or 1.5kV DC) prior to raising the pantograph.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

2F_04.01.Zefiro-Europe.CONCEPT.976	A 4-position switch "Line voltage selector" shall be installed at the driver's desk. With this switch it shall be possible to select the following power supply systems: AC 25 kV 50 Hz, AC 15 kV 16.7 Hz, 3 kV DC, 1.5 kV DC.	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_04.01.Zefiro-Europe.CONCEPT.977	The following positions of the "Line voltage selector" switch shall be connected to TCMS AC 25 kV 50 Hz, AC 15 kV 16.7 Hz, 3 kV DC, 1.5 kV DC.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1069	The voltage selection has to be stable for 1 second before TCMS uses the input signal (only applicable for AC operation).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

In addition country code is needed to identify exactly, in which country the train operates.

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.731	If the train is not designed for operation in all 4 line voltage systems and the driver turns "line voltage selector" to a system which is not implemented, a failure message shall be set and raise of pantographs shall be blocked.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.732	If the driver turns "line voltage selector" to change the selected line voltage system without opening the LCB and lowering the pantograph before, all LCB shall be requested to open and all pantographs shall be lowered immediately.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.700
Description	It shall be ensured that the DC measurement device is not damaged if a DC panto is raised at an AC caternary by mistake.
Safety level	SIL 1

There will be a system detection device at each TT2, T4, T5 and TT7 car. Each will be able to detect as well AC and DC line voltage regardless of what pantograph it is connected to.

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.553	System detection device shall detect and report both AC and DC voltage system.	Derived Requirement	4S_04.-.Power Supply		SIL 2

ID	2F_04.01.Zefiro-Europe.TRS.337
Description	When pantographs are not lowered from the catenary line when passing a neutral section, only those electric circuits on the traction units, which instantaneously conform to the power supply system at the pantograph, may remain connected.
Safety level	SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.707
Description	The high voltage system system shall protect itself against false voltage system selection.
Safety level	SIL 0

When the LCBs and system isolation switches are open, the only equipment that is connected to the line voltage via the pantograph is the system detection device. This device detects if the present line voltage system (AC 25kV, AC 15kV, DC 3kV, DC 1.5kV) is in line with the driver-selected line voltage system.
If present line voltage system does not comply to selected system, pantograph will be lowered and driver gets an indication to select another line voltage system.

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.739	In case the detected system does not fit the selected system – there shall be a message on the IDU – the raise signal shall be canceled and the pantograph become lowered	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.524	If line voltage system is AC and 50Hz is detected propulsion system shall be set up for 25kV 50Hz.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.525	If line voltage system is AC and 16 2/3Hz is detected propulsion system shall be set up for 15kV 16 2/3Hz.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.526	If line voltage system is DC and line voltage is in the range ≥1.975 kV propulsion system shall be set up for 3kV DC.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.527	If line voltage system is DC and line voltage is in the range ≥0.8 kV and <1.975 kV propulsion system shall be set up for 1.5kV DC.	Derived Requirement	4S_04.-.Power Supply		N/A

ID	2F_04.01.Zefiro-Europe.TRS.287
Description	For AC and for DC it shall be detected which line voltage system is connected.
Safety level	N/A

System selection is done automatically by comparing the measured voltage system with the driver-selected voltage system. The system validates the selected voltage system via measurement before allowing the LCBs to be closed. The system detection detects the voltage of contact wire voltage (15kV AC, 25kV AC, 3kV DC or 1.5kV DC).

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.219	The LCB shall be opened resceptively not closed if the system detection detects another line voltage system (15kV AC, 25kV AC, 3kV DC or 1.5kV DC) than selected.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.312	An event shall be set if the line voltage system detected by Propulsion differs from the selected line voltage system by the line voltage selector.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

ID	2F_04.01.Zefiro-Europe.TRS.702
Description	It shall be ensured that the main trafo is not damaged if a AC panto is raised at a DC caternary by mistake.
Safety level	SIL 1

The system detection device will prevent the LCB from closing if the measured system is not the valid voltage system for that LCB.

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.459	It shall not be possible to close the LCB when wrong system is detected compared to the raised pantograph.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.785	If line voltage system sent from Propulsion does not comply to selected system by the line voltage selector the closing of LCB shall be disabled.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.4 Line Circuit Breakers and system isolation switches

3.4.1 Line Circuit Breaker Closing

3.4.1.1 Function Design

3.4.1.1.1 Normal Condition

Connecting of line voltage by closing LCBs.

3.4.1.1.2 Failure Condition

1) LCB does not close on order.
2) LCB closes without order.

3.4.1.1.3 Failure Supervision

1) If the feedback "LCB closed" is not given latest 1s after the "LCB close" order is set, the LCB is considered as faulty. The LCB will be blocked but "activation of line voltage" will retry the closing order.
   After 3 attempts without success within 30 minutes the LCB will be blocked permanently and the pantograph will be blocked.
2) The related pantograph will be lowered, the pantograph disconnector will open and both components will be blocked!

3.4.1.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.468
Description	Line circuit breaker shall be closed when – 'Line circuit breaker close' command given by driver OR automatically when leaving a neutral section AND – pantograph is already lifted AND – no interlocking for maintenance activities active AND – no line circuit breaker opening command The 'Line circuit breaker close' command is done by turning the Line Circuit Breaker switch on driver's desk to position ON.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.978	There shall be 3-position switch for closing/opening of the Line Circuit Breaker. The positions shall be: 1: LCB close (unstable) 2: neutral (stable) 3: LCB open (unstable)	Derived Requirement	4S_08.07.01.-.Drivers Desk		N/A
2F_04.01.Zefiro-Europe.CONCEPT.979	The following positions of the Line Circuit Breaker switch shall be connected to TCMS: 1: LCB close 2: LCB open	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.127	The status of the LCBs shall be displayed on the IDU: – open/closed – faulty (automatic block) – manually blocked	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.970	The LCB shall be inhibited to close if there is no valid country code (country code = 0).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_04.01.Zefiro-Europe.CONCEPT.374	When the driver turns LCB-switch to "close" position AND the pantograph is raised AND line voltage is detected for the selected line voltage system (AC 25kV, AC 15kV, DC 3kV or DC 1.5kV), the LCB which is related to the raised pantograph shall be enabled to close. LCB can be closed from driver's desk only if the master controller is on zero position.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1224	If the LCB is requested (from driver) to close while the Master controller is out of the 0 position, an advisory shall be shown to the driver asking 'to set the master controller in 0 position before ordering LCB closure'	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1090	AC LCB shall only be allowed to close when both AC roof line disconnectors are in end position. In case of plausibility fault of remote AC roof line disconnector, AC LCB shall only be allowed to close when the local disconnector is in open position.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1091	DC LCB shall only be allowed to close when both DC bus bar disconnectors are in end position. In case of plausibility fault of DC roof line disconnector the train bheaviour has to be the following: – in case the fault happens in operation, no action – in case the fault happens at train startup, both roof disconnectors have to be ordered open	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.992	The LCB shall only be allowed to be closed when the LCB enable signal is set from TCMS.	Derived Requirement	4S_04.-.Power Supply		N/A

There are situations when the LCB will remain open even though TCMS is enabling closing of the LCB. The decision to close the LCB will based on internal Propulsion supervision and the enable order from TCMS.

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.500	The system isolation switch shall be closed before the DC LCB can be closed.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.381	When ever the “LCB Close” order is removed, there shall be a 4 second delay before the LCB is allowed to close again.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.386	The number of closings of each LCB shall be stored as condition data.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.532	When the “LCB closed” feedback and “LCB not closed” feedback indicates different states (one indicates closed and one open) for more than 1 second the LCB shall be ordered to open.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.387	When the “LCB closed” feedback and “LCB not closed” feedback indicates different states (one indicates closed and one open) for more than 1 second an event shall be set.	Derived Requirement	4S_04.-.Power Supply	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.531	When the “LCB closed” feedback and “LCB not closed” feedback indicates different states (one indicates closed and one open) for more than 1 second IDU shall show a faulty LCB.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.761	When the feedback "LCB closed" is not given latest 1 second after the "LCB close" order is set, the LCB is considered as faulty. The LCB shall be blocked, but operation of "Close LCB" switch shall retry the closing order. After 3 attempts without success within 30 minutes the LCB will be blocked permanently and the pantograph will be blocked.	Derived Requirement	4S_04.-.Power Supply		N/A

2F_04.01.Zefiro-Europe.CONCEPT.762	If the LCB is closed without order, the related pantograph shall be lowered, the pantograph disconnector shall open and both components shall be blocked!	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1161	After manual exit from parking mode, LCB under selected pantograph has to be closed and LCB under not selected pantograph has to be opened. – In case the new configuration is different from the LCB closed during parking mode and DC line voltage system is selected, the LCB under selected pantograph shall be automatically closed and afterwards the LCB under not selected pantograph shall open. In case in the new configuration the driver has requested raising of both pantographs, this procedure lets closed the LCB which is related to the rear pantograph. – In case the new configuration is different from the LCB closed during parking mode and AC line voltage system is selected, LCB closing command will not be given automatically by TCMS (Driver has to do it manually) This procedure can be only performed at standstill. If the train starts running, this procedure will be stopped immediately opening both LCBs.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1162	While manual exit from parking mode procedure is active, LCB shall close on request from TCMS, if all the other conditions are fullfilled, and it allowed to close both LCBs for a short time only (< 2 s). The LCB which is no longer requested to close from TCMS, shall open without any interruption of the auxiliary power supply.	Derived Requirement	4S_04.-.Power Supply		TBD
2F_04.01.Zefiro-Europe.CONCEPT.1164	The voltage coherence of the two pantographs shall be confirmed, before to allow both LCB to be closed.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.388	In operation mode (refer to key interlocking system), closing of LCB shall be only possible if pantograph is raised and selected line voltage system is confirmed by system detection.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.495	In maintenance mode (refer to key interlocking system), closing of LCB shall be possible if ALL pantographs are lowered.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1177	An event shall be set in case the maintenance mode relays have discrepancy between DM1 and DM8 on the same trainset	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1059	In maintenance mode (refer to key interlocking system), closing of LCB shall be possible if ALL pantographs are lowered.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.796	LCB shall only be allowed to close when converter is disconnected or DC-link is charged to avoid hard switch-on of converters.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.389	When external power supply is active the LCB shall not be allowed to close.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1214	If the driver,with train standtill, request any panto to be rised a warning to the driver shall be set. The warning alerts the driver that at least one 400V external plug is inserted and in case of false reporting the plug can be manually isolated using softkey on iDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.703
Description	Line circuit breaker shall only be allowed to close if valid line voltage is detected
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.347	Line circuit breaker shall only be allowed to close if valid line voltage is detected or train is in maintenance mode. See section 3.4.1.2	Derived Requirement	4S_04.-.Power Supply		SIL 1
2F_04.01.Zefiro-Europe.CONCEPT.1163	During parking mode, in case of missing line voltage the automatic exit procedure starts. In case the line voltage is detected again, the automatic exit procedure shall be stopped and LCB shall be re-closed (only LCB already closed before missing line voltage, no other LCB shall be closed) To avoid simultaneous closing of many trains, LCB shall be automatically closed after a defined amount of time from the line voltage return that is proportional to the train number: time delay in seconds = (train number) + 5, where (train number) is integer in the range 0 to 99, considering the two least significant digits only.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.623
Description	The line circuit breakers of the train shall be closed one after another to limit inrush currents.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.249	If there are 2 trains in multiple the "LCB Close" order shall be delayed 1 second in the second train.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.705
Description	The Line circuit breaker closure shall be synchronised to close at the top of the sinusodial line voltage of 15kV AC. This shall avoid over magnetisation of the main trafo.
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.350	The Line circuit breaker closure shall be synchronised to close at the top of the sinusodial line voltage of 15kV AC. This shall avoid over magnetisation of the main trafo.	Derived Requirement	4S_04.-.Power Supply		N/A

3.4.2 Line Circuit Breaker Opening

3.4.2.1 Function Design

3.4.2.1.1 Normal Condition

Disconnection of Line Voltage will be done by LCB.

3.4.2.1.2 Failure Condition

1) LCB does not open on order.
2) LCB opens without order.

3.4.2.1.3 Failure Supervision

1) The related pantograph will be lowered, the pantograph disconnector will open and both components will be blocked!
2) If the LCB opens without order it will be ordered to close again. If the failure is present 3 times within 30 minutes the LCB will be blocked permanently and the pantograph will be blocked.

3.4.2.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.234
Description	Line circuit breaker shall be opened when – open command from LIM is received OR – line voltage system detection protection or – interlocking for maintenance activities is active OR – drivers emergency push button was pressed OR – ADD on pantograph or – protective shut down from any DCU.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.408	If a line trip is detected the LCB shall be opened.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.750	If a power supply / propulsion line trip is detected the LCB shall be opened.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.409	If the “Emergency push button” is pushed the LCB shall be opened via Train Line (“LCB Emergency Off”).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.751	If the “Emergency push button” is pushed the LCB shall be opened via Train Line (“LCB Emergency Off”).	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1179	If the “Emergency push button” is pushed and the LCB Emergency Off activation is not read in every line trip chain of whole composition an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1071	The LCB shall open immediately, when pantograph down order is given.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.410	The LCB shall open automatically, if any A-Key is activated (key interlocking system).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.752	The LCB shall open automatically, if any A-Key is activated (key interlocking system).	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1180	If any A-Key is activated and the LCB Emergency Off activation is not read in every line trip chain of whole composition an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1181	When IDU soft key is pressed from maintainer, a test of LCB Emergency Off trainline activation shall be carried out (energizing dedicated DOs only in the active cab)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1182	Automatically every time when the LIM test is performed, a test of LCB Emergency Off trainline activation shall be carried out (energizing dedicated DOs only in one cab at the head of composition)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1183	If LCB Emergency Off trainline activation test is performing and the LCB Emergency Off activation is not read in every line trip chain of whole composition an event shall be set	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_04.01.Zefiro-Europe.CONCEPT.1092	When one AC roof line disconnector leaves end position, AC LCB shall be ordered to open.	Derived Requirement	4S_04.-.Power Supply		SIL 1
2F_04.01.Zefiro-Europe.CONCEPT.1093	When one DC bus bar disconnector leaves end position an event shall be set when plausability fault is detected.	Derived Requirement	4S_04.-.Power Supply		SIL 1
2F_04.01.Zefiro-Europe.CONCEPT.763	When LCB does not open on order, the related pantograph shall be lowered, the pantograph disconnector shall open and both components shall be blocked!	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

2F_04.01.Zefiro-Europe.CONCEPT.764	When the LCB opens without order, it shall be ordered to close again. If the failure is present 3 times within 30 minutes the LCB will be blocked permanently and the pantograph shall be blocked.	Derived Requirement	4S_04.-.Power Supply		N/A

ID	2F_04.01.Zefiro-Europe.TRS.704
Description	Line circuit breaker shall be opened when – 'Line circuit breaker open' command given by driver OR – entering neutral section
Safety level	SIL 0

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_04.01.Zefiro-Europe.CONCEPT.353	When the Line Circuit Breaker switch is turned to “LCB open” the LCB shall be ordered to open (also in rear train in case of multiple operation).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1056	LCB shall open when country code becomes invalid (country code = 0).	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.403	A permanently blocked (due to a failure) LCB shall be blocked until a master reset is made or PCU is restarted.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.995	If LCB is permanently blocked (due to failure), related pantograph shall be ordered to lower and related pantograph disconnector shall open and both, pantograph and disconnector, shall be blocked.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.406	If the feedback ‘LCB Closed’ changes to ‘LCB not closed’ while the ‘LCB close' order is high, the status of the LCB is inconsistent and the LCB shall be ordered to open and the LCB shall be permanently blocked.	Derived Requirement	4S_04.-.Power Supply		SIL 1
2F_04.01.Zefiro-Europe.CONCEPT.537	If the feedback ‘LCB Closed’ changes to ‘LCB not closed’ while the ‘LCB Close' order is high, the status of the LCB is inconsistent and an event shall be set.	Derived Requirement	4S_04.-.Power Supply	EVENT	N/A

<table>
<tr><td>2F_04.01.Zefiro-Europe.CONCEPT.986</td><td>If all LCBs in the trains are open (valid also for Multiple Unit), the 'LCB is open' indication lamp shall be illuminated on driver's desk.

For more details see the tables below for multiple unit and single unit:

single unit:
<table>
<tr><td colspan="3">SINGOLO</td></tr>
<tr><td>IR1</td><td>IR2</td><td>spia</td></tr>
<tr><td>0</td><td>0</td><td>accesa</td></tr>
<tr><td>0</td><td>1</td><td>spenta</td></tr>
<tr><td>1</td><td>0</td><td>spenta</td></tr>
<tr><td>1</td><td>1</td><td>spenta</td></tr>
</table>

multiple unit:
<table>
<tr><td>IR1</td><td>IR2</td><td>IR3</td><td>IR4</td><td>spia</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>accesa</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>1</td><td>LAMP</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>LAMP</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>1</td><td>LAMP</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>LAMP</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>1</td><td>spenta</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>spenta</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>1</td><td>spenta</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>LAMP</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>1</td><td>spenta</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>0</td><td>spenta</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>spenta</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>0</td><td>LAMP</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>spenta</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>0</td><td>spenta</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>spenta</td></tr>
</table>

0 = aperto
1 = chiuso</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 0</td></tr>
<tr><td>2F_04.01.Zefiro-Europe.CONCEPT.985</td><td>When ordered by TCMS, the indication lamp 'LCB is open' shall be illuminated on the driver's desk.</td><td>Derived Requirement</td><td>4S_09.02.09.-.Conventional Train Control - Circuits</td><td>HW HMI</td><td>SIL 0</td></tr>
<tr><td>2F_04.01.Zefiro-Europe.CONCEPT.727</td><td>If the cab is deactivated without entering the parking mode, all LCB shall be reqested to open.</td><td>Derived Requirement</td><td>4S_09.03.05.-.TCMS Software</td><td></td><td>SIL 0</td></tr>
</table>

ID	2F_04.01.Zefiro-Europe.TRS.712
Description	Line circuit breaker shall be opened fast enough when system change section is entered under speed to avoid damage of electric. equipment
Safety level	SIL 1

If the train is passing a system separation section with a manual indication in proper time or a correct signal from external sensors, the LCB will open before entering the system separation section. See section 3.3.4

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_04.01.Zefiro-Europe.CONCEPT.464	If line voltage lower than 1.8kV in DC 3kV or 0.85kV in DC 1.5kV for more than 1 second when the train speed is >30km/h the pantograph shall be requested to lower directly after the LCB has opened.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1105	The pantograph shall be lowered using the pantograph fast down trainline when requested from PCU (propulsion control unit).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1070	The pantograph fast down trainline shall be set when ordered by TCMS DO.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_04.01.Zefiro-Europe.TRS.714
Description	Line circuit breaker shall open in case of risk for overtemperature for trafo
Safety level	SIL 1

Pre-Condition: Temperature measurement has to be installed at main transformer.

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.765	If an over temperature in main transformer at AC operation or line inductor at DC operation is detected, the related LCB shall open and an EVENT shall be set.	Derived Requirement	4S_04.-.Power Supply	EVENT	SIL 1
2F_04.01.Zefiro-Europe.CONCEPT.497	If an over temperature in the transformer is detected running AC for TBD time in TBD minutes the high voltage section (with the over heated transformer) shall be isolated in AC operation (refer to section 3.4.3 “Isolate high voltage”). PPC to define TBDs.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.498	If an over temperature in line inductor is detected running DC for TBD time in TBD minutes the high voltage section (with the over heated line inductor) shall be isolated in DC operation (refer to section 3.4.3 “Isolate high voltage”). PPC to define TBDs.	Derived Requirement	4S_04.-.Power Supply		N/A

ID	2F_04.01.Zefiro-Europe.TRS.624
Description	In case of normal opening (no protection situation) the Line Converter Modules shall be ordered to ramp down before opening the line circuit breaker
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.252	Any order of normal opening of LCB requires a ramp down of line converters before LCB can be opened.	Derived Requirement	4S_04.-.Power Supply		N/A

ID	2F_04.01.Zefiro-Europe.TRS.326
Description	In case of fire, the power supply to the affected system shall be interrupted. If one traction circuit is concerned, the respective line circuit breaker shall open, the affected system isolated from the traction circuit and the line circuit breaker then close again.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.255	If fire is detected in the underframe the LCB shall be requested to open and the disconnectors shall be requested to open.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.465	If the LCB and roof line disconnectors has opened due to fire, this half of the train where the fire is detected shall be isolated (disable closing of LCB and block raising of pantograph) and an event shall be set to inform the driver that change of pantograph is required.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1100	It shall be possible to manually cut out a LCB.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1101	If the LCB is manually cut out the LCB an event shall be set as long as the LCB is cut out.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

2F_04.01.Zefiro-Europe.CONCEPT.1102	A LCB that is manually cut out shall be opened and not closed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

After 1 sec. that any AC LCB is open in the consist (single or multiple operation) and a voltage of AC is detected, an even shall be set and the related panto must be blocked.
The line voltage to take in consideration shall be the one after the AC LCB that is provided with the variable

PCU -> CCU-O	Signal	Cycle time	Description	Scale
CX1CT607	HCV_XU_AcLn	64 ms	LCM AC line voltage to remote system	1=12V
CX1CT6C2.7	HCV_VU_AcLn	64 ms	Valid bit: LCM AC line voltage to remote system	L

The function is activated if the voltage is higher than 15 kV (under 25 kv operation) after 1 sec. even if any LCB is in open position, the panto related to the LCB that was in closed position before must be put in fault.
An event to the driver shall be set with the following sentence" LCB X non aperto, il pantografo y e inibito e non deve essere effettuato nessun master reset". Also the buzzer shall be activated.
This logic must be done on consist level, this means that if an AC LCB on the rear/front train is not opened, the massage must be send to the driver in the front train where the cab is active.
The message for the diver are:
For the fault of the LCB in T4
"Abbassare l''interruttore automatico GB-F006 posto nell' armadio 59. C1 del veicolo T4"

For the fault of the LCB in T5
"Abbassare l''interruttore automatico GB-F006 posto nell' armadio 59. C1 del veicolo T5"

3.4.3 Isolate High Voltage

3.4.3.1 Function Design

3.4.3.1.1 Normal Condition

It is possible to isolate short circuits or failures to keep half of the traction of an 8 car train available.

3.4.3.1.2 Failure Condition

1) One disconnector does not open on order.
2) One disconnector does not close on order.
3) Position of disconnectors not known.

3.4.3.1.3 Failure Supervision

1) Roof line disconnector does not open on order. Event will be set and driver must change pantograph manually.
2) Roof line disconnector does not close on order. Event will be set and driver must change pantograph manually.
3) Roof line disconnector is in an unknown position. Event will be set and driver must change pantograph manually.

3.4.3.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.474
Description	It shall be possible to isolate separate high voltage sections.
Safety level	SIL 0

A train half is considered "isolated from DC catenary" when the DC line disconnector related to active pantograph is open and either its DC pantographs are lowered or the system isolating switch is open.

A train half is considered "isolated from AC catenary" when the AC line disconnector related to active pantograph is open and either its AC pantographs are lowered or the AC LCB is open.

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.997	The roof line disconnector shall be considered to be not opening on order if both feedbacks indicate closed position after: – 3 seconds for AC disconnector – 1 second for DC disconnector Three attempts shall be allowed to open with 10 seconds interval between each attempt. If after 3 attempts the roof line disconnector has not opened on order the roof line disconnector shall be indicated as faulty to TCMS.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.998	The roof line disconnector shall be considered to be not closing on order if both feedbacks indicating open position after: – 3 seconds for AC disconnector – 1 second for DC disconnector Three attempts shall be allowed to close with 10 seconds interval between each attempt. If after 3 attempts the disconnector has not closed on order then both disconnectors shall be ordered to open.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.999	The roof line disconnector shall be considered to be in an unknown position if neither open or close acknowledge is set or open and close acknowledge are set simultaneously for 4 seconds. Three attempts shall be allowed to open with 10 seconds interval between each attempt. If the position is still unknown after 3 attempts roof line disconnectors shall be indicated as faulty to TCMS.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1000	If the roof line disconnector (only in AC mode) is indicated as faulty from the PCU (propulsion control unit) then the pantograph related to the faulty roof line disconnector shall be blocked and an event shall be set to indicate to the driver that pantograph change is required. Note: the bheaviour here above is valid only for AC mode	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1222	If the system isolation switch is indicated as faulty from the PCU (propulsion control unit) then the pantograph related to the faulty switch shall be blocked and an event shall be set to indicate to the driver that pantograph change is required.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.474	The roof line disconnectors shall be closed at startup of train	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1223	The TCMS, since the start up of the train till the shut down, has to send the status of the battery contactor closed to the propulsion system.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.475	The status (open/closed) of the roof line disconnectors shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.137	If a failure in HV-part is detected, which requires separation of both halves of the train (e.g. earth fault), the roof line disconnector related to active pantograph shall be requested to open.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.467	The DC-line disconnectors shall only be operated when all pantographs are lowered or DC LCBs are opened.	Derived Requirement	4S_04.-.Power Supply		N/A

2F_04.01.Zefiro-Europe.CONCEPT.586	The AC-line disconnector shall only be operated when all AC pantographs are lowered or all AC LCBs are in open position.	Derived Requirement	4S_04.-.Power Supply		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1226	the IDU has to provide a cut in/out virtual button for the driver to request the manual opening of the the roof disconnectors 3 or 25 kV. This button must be slevtive for train and line voltage.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	
2F_04.01.Zefiro-Europe.CONCEPT.1227	When the TCMS recives the roof disconnector cut out request, the cut out request has to be sent to propulsion only if: – Train is standstill AND – All panto are down OR All LCB are open	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1228	When the roof disconnector cut out request is ordered only the selected voltage (TCMS has to check the driver desk selector) roof disconnector shall be opened. Note: if 3kv is selected and the driver request the roof disconnector opening only the DC disconnector request has to be sent to propulsion (see icd variable for request)	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1231	When the roof disconnector opening request is ordered from the driver the TCMS implements the following actions: – a Master reset is sent to all equipments – 10 s after the master reset the TCMS request to propulsion to open both (3 kV or 25kV) roof disconnector	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1229	In case the roof disconnector opening order is triggered to propulsion an event shall be set for enginering	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	
2F_04.01.Zefiro-Europe.CONCEPT.1230	In case the the driver request a roof disconnector opening but not all conditions are true a selective (one event for each condition not verified) event shall be set to the driver	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	
2F_04.01.Zefiro-Europe.CONCEPT.1232	The roof disconnectors can be cut in if: – a cot out command has been previously delivered AND – Train is standstill AND – All panto are down AND – All LCB are open	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1233	In case the roof disconnector closing order is triggered to propulsion an event shall be set for enginering	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	
2F_04.01.Zefiro-Europe.CONCEPT.1234	In case the the driver request a roof disconnector clossing but not all conditions are true a selective (one event for each condition not verified) event shall be set to the driver	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	

3.5 Key interlocking

The Key Interlocking System is described in the document 3EST000226-2279.

3.5.1 Function Design

3.5.1.1 Normal Condition

All power supply available: HV, 400V, 110V

3.5.1.2 Failure Condition

Failure conditions and failure effects are described in the FMECA-File “FMECA KIS Zefiro Italy_XXX.xls”.

3.5.1.3 Failure Supervision

Status of HV-switches and disconnectors is controlled by TCMS and hardwired to control lamp and solenoid protected lock.

3.5.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.679
Description	Two levels of key interlocking shall be supported: – Operation mode – Maintenance mode
Safety level	SIL 2

At Interlocking sequence there are five different modes defined:

- Operation Mode
- Parking Mode
- Disabled Mode
- Depot Supply Mode
- Maintenance Mode

ID	2F_04.01.Zefiro-Europe.TRS.680
Description	In operation mode, high voltage components and auxiliary power supply shall be locked to prevent unwanted and dangerous switching operations.
Safety level	SIL 2

ID	2F_04.01.Zefiro-Europe.TRS.681
Description	Changing from “operation” to “maintenance” mode, the following steps shall be performed: – open line circuit breakers – lowering pantographs – earthing the high voltage equipment, if necessary
Safety level	N/A

1) Operation Mode: HV is live, active MCB is closed, active pantograph is raised, internal 3AC400V is live, 110 VDC is live, depot supply is not possible. KIS is not active.

2a) Parking Mode: is identical with Operation Mode, but the **traction is blocked and the Parking brake is applied**.

2b) Disabled Mode: **HV shut off, all MCBs are open, all pantographs are down, both Isolating/Grounding Switches DC (4a) are in earthed position, depot supply is possible, contactors for 3AC400V are open,** and 110 VDC remains live.
Disabled Mode is a parking mode for the train with the possibility to connect depot supply; e.g. for cleaning the train with a vacuum cleaner or charging the 110V batteries. To achieve this mode, the KIS is not active.

3) Depot Supply Mode: HV shut off **and earthed**, all pantographs are down, depot supply is possible, 110 VDC remains live for a limited time; **access to the HV boxes at the roof is possible**.
   Depot Supply Mode is a mode, where maintenance work can be performed at cubicles containing live 3AC400V equipment. Also maintenance at roof equipment is possible. To achieve this mode, the KIS is needed.

4) Maintenance Mode: HV shut off and earthed, pantographs are down, **3AC400 V is shut off and earthed, DC-links are earthed; access to the key-locked cubicles is possible**, 110 VDC remains live, **depot supply is not possible**.
   Maintenance Mode is a mode, where maintenance work can be performed at cubicles containing HV equipment or for exchanging 3AC400V equipment. To achieve this mode, the KIS is needed.

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.602	If any A-Key is activated the LCB shall open and the pantograph shall lower	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_04.01.Zefiro-Europe.TRS.682
Description	The hatches of the converter unit must be locked to prevent access to dangerous voltages.
Safety level	SIL 2

3.5.3 Function Earthing of High Voltage Supply

3.5.3.1 Function Design

3.5.3.1.1 Normal Condition

The High Voltage system is earthed to secure that maintenance of the system can be done in a secure way without risk for electric shock.

3.5.3.1.2 Failure Condition

Failure conditions and failure effects are described in the FMECA-File “FMECA KIS Zefiro Italy_XXX.xls”.

1) Isolating/Grounding Switch DC (4a) does not close
2) Isolating/Grounding Switch AC (4b) does not close

3.5.3.1.3 Failure Supervision

1) HV DC line is earthed manually by HV DC Earthing Disconnectors 36a and 36b.
2) Status of Isolating/Grounding Switch AC is controlled by PCU (via TCMS) and hardwired to a control lamp and a solenoid protected lock

3.5.3.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.291
Description	It shall be possible to earth the high voltage supply with a switch remotely controlled. Earthing of the high voltage AC supply shall be done after the line circuit breaker. With this measure the main transformer shall also be earthed.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1016	When the push-button “Lamp Test” from the A1-key Panel in DM1-car is pressed the Blue lamp “HV AC earthed” at A1-Key-Panel will light.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1017	When the push-button “Lamp Test” from the A2-key Panel in DM8-car is pressed the Blue lamp “HV AC earthed” at A2-Key-Panel will light.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

2F_04.01.Zefiro-Europe.CONCEPT.147	When the Key from the A-key Panel in one DM-car is moved to the Depot- Supply position the AC- and DC-LCBs shall be ordered to open and all pantographs shall be lowered.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1060	When the Key from the A-key Panel in one DM-car is moved to the Depot- Supply position the AC- and DC-LCBs shall be ordered to open and all pantographs shall be lowered.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1018	When the Key from the A-key Panel in one DM-car is moved to the Depot- Supply position all 3AC400V contactors 18a, 18c and 39 shall open.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1078	When the Key from the A-key Panel in one DM-car is moved to the Depot- Supply position all 3AC400V contactors 18a shall open.	Derived Requirement	4S_05.-.Propulsion		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1218	when the A-key is restored to 'normal mode' position (from 'Depot mode') all MV contactors 18c (medium voltage contactor 7) and 39 (medium voltage contactor 6) and also the medium voltage contactor 8 shall be closed.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1040	When the Key from the A-key Panel in one DM-car is moved to the Depot- Supply position pre charging devices 29a, 29b, 32a, 32b and disconnectors 33a, 33b shall open.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1079	When the Key from the A-key Panel in one DM-car is moved to the Depot- Supply position pre charging devices 29a, 29b, 32a, 32b and disconnectors 33a, 33b shall open.	Derived Requirement	4S_05.-.Propulsion		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1019	When the A1-Key at A1-Key Panel in DM1-car or the A2-Key at A2-Key Panel in DM8-car is moved to the Depot-Supply position the Grounding switches 4a (TT2 and TT7-car) and 4b (T4 and T5-car) shall be ordered to close the earthing contacts.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1080	When the A1-Key at A1-Key Panel in DM1-car or the A2-Key at A2-Key Panel in DM8-car is moved to the Depot-Supply position the Grounding switches 4a (TT2 and TT7-car) and 4b (T4 and T5-car) shall be ordered to close the earthing contacts.	Derived Requirement	4S_04.-.Power Supply		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1020	When the A1-Key or the A2-Key is moved to the Depot-Supply position the Blue lamps “HV AC earthed” at both A-Key-Panels in DM1- and DM8-car will light, if Grounding switches AC 4b in T4- and T5-car are in the earthed position.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1021	When the A1-Key or the A2-Key is moved to the Depot-Supply position the solenoid protected locks for the A-Keys at HV DC Earthing Disconnector 36a and 36b (TT2- and TT7-car) will be unlocked, if Grounding switches AC 4b in T4- and T5-car are in the earthed position.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.413	The status of the Grounding switches 4a and 4b shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.414	All orders for earthing shall be given simultaneously to all HV Grounding switches 4a and 4b on the train.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.417	The close order signal shall be set low as soon as the feedback Grounding switch 4a and 4b CLOSED goes high.	Derived Requirement	4S_04.-.Power Supply		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.419	The open order signal shall be set low as soon as the feedback Grounding switch 4a and 4b OPEN goes high.	Derived Requirement	4S_04.-.Power Supply		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.420	The maximum time for close or open order shall be 4 seconds.	Derived Requirement	4S_04.-.Power Supply		N/A

2F_04.01.Zefiro-Europe.CONCEPT.421	If the A-key has not been turned to the Depot - Supply position for >7days, then there shall be an event displayed on the IDU. The event shall be reset when the key has been in the Depot - Supply position and the Grounding switches 4a (open when earthed) and 4b (closed when earthed) have followed the right sequence when the A-key is returned to the normal position (4a remains open till pantograph command, 4b is open when not earthed). This ensures that the key is operated to test the Grounding switches function at least every 7 days.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.422	The Grounding Switches 4b shall not be able to close when any pantograph is raised.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.755	The Grounding Switch 4b shall not be ordered to close when any pantograph is raised.	Derived Requirement	4S_04.-.Power Supply		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.412	When the input from the A-key panel in both DM-cars is not in the Depot- Supply position the Grounding switches 4b (T4 and T5-car) shall be ordered to open.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Earth HV DC line manually (left side of the train):
Actions: Manual actions

- Remove A1-key from A1-key panel and put it into the lock of HV DC Earthing Disconnector panel switch 36a (Z-switches in car TT2) and turn it.
- Turn 36a (Z-switches) into earthing position

Results: mechanical interlocking

- The left DC-HV box is earthed by disconnector 36a (Z-switches) manually
- One E1-keys is released for turning and taking off; mechanically
- One B1-key is released for turning and taking off; mechanically

The A1-key cannot be removed if the switch 36a is in earthed position; mechanically

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1024	The position of HV DC Earthing Disconnector 36a (Z-switches) shall be monitored by TCMS with two auxiliary contacts.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

Earth HV DC line manually (right side of the train):
Actions: Manual actions

- Remove A2-key from A2-key panel and put it into the lock of HV DC Earthing Disconnector panel switch 36b (Z-switches in car TT7) and turn it.
- Turn 36b (Z-switches) into earthing position

Results: mechanical interlocking

- The right DC-HV box is earthed by disconnector 36b (Z-switches) manually
- One E2-keys is released for turning and taking off; mechanically
- One B2-key is released for turning and taking off; mechanically

The A2-key cannot be removed if the switch 36b is in earthed position; mechanically

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1026	The position of HV DC Earthing Disconnector 36b (Z-switches) shall be monitored by TCMS with two auxiliary contacts.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

3.5.4 Function Earthing of Converter boxes

3.5.4.1 Function Design

3.5.4.1.1 Normal Condition

The High Voltage system is earthed to secure that maintenance of the system can be done in a secure way without risk for electric shock.

3.5.4.1.2 Failure Condition

Failure conditions and failure effects are described in the FMECA-File “FMECA KIS Zefiro Italy_XXX.xls”.

1) The earthing switch does not close on order, that is the system is not earthed

3.5.4.1.3 Failure Supervision

1) Key to converter unit can not be released if the earthing switch is not in earthed position.

3.5.4.2 Requirements

ID	2F_04.01.Zefiro-Europe.TRS.683
Description	An earth switch shall be installed near each converter unit. The earth switch is operated to earth the internal connections (DC-link) of the converter unit in maintenance mode. In order to access the converter unit hatches, the earth switch shall be operated, which gives access to a key to the hatch.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.151	An earth switch shall be installed on the DC link of the converter to discharge all capacitors before the maintenance on the converter units.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1054	**Blocking 3AC400V (right side and left side of the train):** Actions: manual action • Turn B1-key at the HV DC Earthing Disconnector panel 36a Results: • Blocking 3AC400V will be initiated by opening 18b (Depot Supply Contactor) **on both sides of the train**. The B1-key can be released. mechanically	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_04.01.Zefiro-Europe.CONCEPT.1055	**Blocking 3AC400V (right side and left side of the train):** Actions: manual action • Turn B2-key at the HV DC Earthing Disconnector panel 36b Results: • Blocking 3AC400V will be initiated by opening 18b (Depot Supply Contactor) **on both sides of the train**. The B2-key can be released. mechanically	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

Earthing 3AC400V (left side of the train):
Actions: manual action

- take B1-key, put it into the 400V Earthing Disconnector 37a of car T4, and turn it.
- Turn handle of 37a and earth.

Results:

- The B1-key cannot be removed; mechanically
- two C1-keys will be released; mechanically

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1030	The position of 400V Earthing Disconnector 37a shall be monitored by TCMS with two auxiliary contacts.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1081	When the 400V Earthing Disconnector is closed it shall be displayed on IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

Earthing 3AC400V (right side of the train):
Actions: manual action

- take B2-key, put it into the 400V Earthing Disconnector 37b of car T5, and turn it.
- Turn handle of 37b and earth.

Results:

- The B2-key cannot be removed; mechanically
- two C2-keys will be released; mechanically

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1031	The position of 400V Earthing Disconnector 37b shall be monitored by TCMS with two auxiliary contacts.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_04.01.Zefiro-Europe.CONCEPT.1057	When the 400V Earthing Disconnector is closed it shall be displayed on IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

Earthing the DC-links (left side, cars DM1 and M3):
Actions: manual action

- Take the two C1-keys, put them into the locks of the two DC-Link Earthing Disconnectors 35a (cars DM1 and M3), and turn them.
- Turn handles of the earthing disconnectors and earth.

Results:

- DC links and HV path (step down chopper (11)) are earthed.
- The C1-keys cannot be released; mechanically
- Two D1-keys from car DM1 and two D2-keys from car M3 are released; mechanically

Earthing the DC-links (right side, cars DM8 and M6):
Actions: manual action

- Take the two C2-keys, put them into the locks of the two DC-Link Earthing Disconnectors 35b (cars DM8 and M6), and turn them.
- Turn handles of the earthing disconnectors and earth.

Results:

- DC links and HV path (step down chopper (11)) are earthed.
- The C2-keys cannot be released; mechanically
- Two D3-keys from car DM8 and two D4-keys from car M6 are released; mechanically

Getting the F-Keys (left side)
Actions: left side, at car DM1; manual action

- Take one of the D1-, D2-, D3- and D4-keys and put all four keys into the F Key Multiplier at car DM1
- Turn them all

Result:

- 12 F-keys are released
- The D1-, D2-, D3- and D4-keys cannot be released, if at least one F-key is taken off.

Getting the F-Keys (right side)
Actions: right side, at car DM8; manual action

- Take one of the D1-, D2-, D3- and D4-keys and put all four keys into the F Key Multiplier at car DM8
- Turn them all

Result:

- 12 F-keys are released
- The D1-, D2-, D3- and D4-keys cannot be released, if at least one F-key is taken off.

Bringing back the train into Operation Mode
The sequence of the KIS has to be performed into the reverse direction for each side of the train.
First all hatches protected by the F-keys have to be closed. The F-keys have to be taken back into its key multipliers in order to get access to the D1-, D2-, D3 and D4-keys. (mechanically)
The D1-, D2-, D3- and D4-keys have to be brought back into the DC-Link Earthing Disconnectors 35a+b, the earthing handles have to be brought into the "not earthed" position and then the C1- and C2-keys are released. (manually)
With the two C1-keys (and the two C2-keys) the 400V Earthing Disconnectors 37a+b for 3AC400V are reset by switching into the "not earthed" position. So, the B1- and B2-keys are released. (manually)
The B1/B2-keys and the E1/E2-keys have to be brought back into the disconnector panels 36a+b and turned, and then switches 36a+b must be turned into the “not earthed” position which will release the A1 and A2-keys. (manually)
Now we are back in Operation Mode.

The A1/A2-keys must be brought back into the A-key panels 40a/b and turned. This opens the Grounding switches 4b from AC-MCBs in cars T4 and T5.

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1038	After opening of the Grounding switches 4b (left and right AC-MCB) the solenoids in the HV DC Earthing Disconnectors 36a and 36b must switched off and the lock from A1-Key and A2-Key is blocked.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_04.01.Zefiro-Europe.CONCEPT.1039	After opening of the Grounding switches 4b (left and right AC-MCB) the Blue lamps “AC HV earthed” at A1- and A2-Key-Panel will switch off.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

Now the Disabled Mode is achieved.
To bring the train into Parking Mode the Tumbler Switches “pantograph up” and “MCB closed” have to operated.
From Parking Mode the train can be brought back into Operation Mode by releasing the parking brakes and releasing the traction control.

3.6 HVAC pre-conditioning on DC line

A Pre-Conditioning (PC) functionality shall be foreseen on DC lines in order to allow train to supply all the needed power to HVAC system without exceeding current limitation (200 Amps per pantograph on 3kVdc, 300 Amps per pantograph on 1.5Vdc networks)
Pre conditioning functionality will be requested to HVAC under DC line voltage only when requested by the driver. On AC lines instead, PC mode will be automatically managed by TCMS

3.6.1 Requirements

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1192	A dedicated menu on IDU shall be foreseen to enable Pre-Conditioning function	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	
2F_04.01.Zefiro-Europe.CONCEPT.1193	TCMS shall check that all following condition are true before let the driver enable PC mode: – Train standstill – Parking brake applied – Any DC system is selected (3kV or 1.5kV) – Both pantographs are down (and all LCB open) – NO panto or LCB are set in faulty/cut-out status – NO ACM are set in faulty/cut-out status – NO Towing mode is active – NO Fire alarm active on the train – Propulsion system ready (*"Status: Preheating possible"* signal is true)	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1194	PCU shall provide to TCMS interface signal to manage pre-conditioning mode – *Status: Preheating possible* – *Command: Activate Preheating*	Derived Requirement	4S_05.-.Propulsion		
2F_04.01.Zefiro-Europe.CONCEPT.1195	If all following condition are not verified, PCU shall set true *"Status: Preheating possible"* signal: – No communication with the LIM – No communication with remote high voltage – DC system is isolated – DC pantograph is isolated – Any DC system disconnector failure is active – Line trip global relay cannot open (all other line trip related failures are covered above	Derived Requirement	4S_05.-.Propulsion		
2F_04.01.Zefiro-Europe.CONCEPT.1196	If Driver has pressed PC soft key on IDU, TCMS shall: – set *"Command: Activate Preheating"* to all leader PCUs – activate traction block – Block panto raising command till both DC roof line disconnector are open	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1197	When *"Command: Activate Preheating"* signal is set, leader PCU shall open both DC line roof disconnector and configure Line Trip chain in Local mode	Derived Requirement	4S_05.-.Propulsion		
2F_04.01.Zefiro-Europe.CONCEPT.1198	In case one or both panto does not raise on order a warning to the driver shall be displayed that PC is not possible	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	
2F_04.01.Zefiro-Europe.CONCEPT.1199	In PC mode, with both panto up (and roof disconnector open) when driver commands LCB closure, order shall be set in both half train	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1200	In case one or both LCB does not close on order a warning to the driver shall be displayed that PC is not possible	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	

2F_04.01.Zefiro-Europe.CONCEPT.1201	When all ACM have started (in operation), TCMS shall activate *“Preconditioning mode request”* in all HVAC units	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1202	If within 60 sec from LCB closure, ALL ACM are not in operation status, LCB shall be opened by TCMS and a warning displayed to the driver (driver is free to choose if re-close LCB and try again or abort PC mode definitively)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	
2F_04.01.Zefiro-Europe.CONCEPT.1203	If AC mode is selected, pre-conditioning mode has to be managed automatically by TCMS (according HVAC SFRS - 2F_15.02.Zefiro-Europe.TRS.61)	Derived Requirement	4S_09.03.05.-.TCMS Software		

Abort pre-conditioning mode

Nr	Description	Type	Derived to	User Interface	SIL
2F_04.01.Zefiro-Europe.CONCEPT.1204	An advice “preconditioning finished” shall be displayed to the driver when all HVAC go out of *“Preconditioning”* status	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	
2F_04.01.Zefiro-Europe.CONCEPT.1205	If driver presses Pre-conditioning softkey, when PC mode is active, *“Preconditioning mode”* shall be reset to HVAC (*“Automatic mode”* will be enabled), then request to open LCB and lower panto has to be displayed on IDU	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1206	When both panto are lowered after driver stop PC mode via IDU, TCMS will reset *“Command: Activate Preheating”*	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1207	When *“Command: Activate Preheating”* signal is reset by TCMS, leader PCU shall close both DC line roof disconnector and reset Line trip chain configuration	Derived Requirement	4S_05.-.Propulsion		
2F_04.01.Zefiro-Europe.CONCEPT.1208	In case any precondition is missing (except *“Both pantographs are down”*), TCMS shall – open LCB and reset *“Preconditioning mode request”* signal to all HVAC units – display an warning to the driver that pre-conditioning is not possible – after LCB is open request to lower pantographs shall be displayed to the driver and when driver lowers panto *”Command: Activate Preheating”* shall be set LOW to leader PCUs	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_04.01.Zefiro-Europe.CONCEPT.1209	In case during PC mode, one (or both) LCB is opened by local protection (but Status: *Preheating possible* signal is still set), second LCB shall be opened by TCMS and a warning displayed to the driver (driver is free to choose if re-close LCB and try again or abort PC mode definitively)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	

2F_04.01.Zefiro-Europe.CONCEPT.1210	In case during PC mode, Line Voltage ≤ 2,2 kV DC on 3kV lines (or ≤1,4kV on 1,5kV lines), if it doesn't rise over the threshold within 30s TCMS shall open all LCB (all parameters, time and voltage values, shall be tunable during commissioning) When line voltage comes back and it is over the threshold for more than 30s, LCB shall be automatically closed and pre-conditioning can start again (TCMS shall wait that ALL ACM are in operation and then sent *"Preconditioning mode request"* to all HVAC) (all parameters, time and voltage values, shall be tunable during commissioning) TCMS shall set a warning to the driver describing that pre-conditioning is not possible	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	
2F_04.01.Zefiro-Europe.CONCEPT.1211	In case during PC mode, 1 ACM is stopped/faulty and it does not restart within 30s, allLCB shall be opened by TCMS and a warning displayed to the driver (driver is free to choose if re-close LCB and try again or abort PC mode definitively)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	
2F_04.01.Zefiro-Europe.CONCEPT.1212	In case during PC mode, at least 2 ACM are stopped/faulty, all LCB shall be immediately opened by TCMS and a warning displayed to the driver (driver is free to choose if re-close LCB and try again or abort PC mode definitively)	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	

4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:
Outputs:

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.30	Pantograph Raised	L	2F_04.01.Zefiro-Europe.CONCEPT.284 2F_04.01.Zefiro-Europe.CONCEPT.309 2F_04.01.Zefiro-Europe.CONCEPT.310 2F_04.01.Zefiro-Europe.CONCEPT.484
2F_04.01.Zefiro-Europe.CONCEPT.587	Pantograph Lowered	L	2F_04.01.Zefiro-Europe.CONCEPT.207 2F_04.01.Zefiro-Europe.CONCEPT.284
2F_04.01.Zefiro-Europe.CONCEPT.590	Pantograph disconnector open	L	2F_04.01.Zefiro-Europe.CONCEPT.566 2F_04.01.Zefiro-Europe.CONCEPT.567
2F_04.01.Zefiro-Europe.CONCEPT.1077	Pantograph disconnector closed		2F_04.01.Zefiro-Europe.CONCEPT.566 2F_04.01.Zefiro-Europe.CONCEPT.567
2F_04.01.Zefiro-Europe.CONCEPT.604	A-key activated	L	2F_04.01.Zefiro-Europe.CONCEPT.602
2F_04.01.Zefiro-Europe.CONCEPT.1044	Line voltage selector: AC 25 kV 50 Hz		2F_04.01.Zefiro-Europe.CONCEPT.492
2F_04.01.Zefiro-Europe.CONCEPT.1045	Line voltage selector: AC 15 kV 16.7 Hz		2F_04.01.Zefiro-Europe.CONCEPT.492
2F_04.01.Zefiro-Europe.CONCEPT.1046	Line voltage selector: 3 kV DC		2F_04.01.Zefiro-Europe.CONCEPT.492
2F_04.01.Zefiro-Europe.CONCEPT.1047	Line voltage selector: 1.5 kV DC		2F_04.01.Zefiro-Europe.CONCEPT.492
2F_04.01.Zefiro-Europe.CONCEPT.1048	LCB switch: LCB close		2F_04.01.Zefiro-Europe.CONCEPT.374
2F_04.01.Zefiro-Europe.CONCEPT.1049	LCB switch: LCB open		2F_04.01.Zefiro-Europe.CONCEPT.374

PCU major fault

Outputs:
Pantograph raise order
Pantograph raise order backup mode

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.597	Fast lowering of all pantographs	L	2F_04.01.Zefiro-Europe.CONCEPT.296
2F_04.01.Zefiro-Europe.CONCEPT.589	Indication lamp: LCB is open	L	2F_04.01.Zefiro-Europe.CONCEPT.985 2F_04.01.Zefiro-Europe.CONCEPT.986

2F_04.01.Zefiro-Europe.CONCEPT.1075	Command: Close pantograph disconnector		2F_04.01.Zefiro-Europe.CONCEPT.563
2F_04.01.Zefiro-Europe.CONCEPT.1076	Command: Open pantograph disconnector		2F_04.01.Zefiro-Europe.CONCEPT.565

4.2 Bus Interface

Supplier to TCMS SW: PCU -> CCU-O

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.901	Command: enable start of remote 1500V DC system		
2F_04.01.Zefiro-Europe.CONCEPT.902	Command: enable start of 1500V DC system		
2F_04.01.Zefiro-Europe.CONCEPT.903	Command: enable start of remote 3kV DC system		
2F_04.01.Zefiro-Europe.CONCEPT.904	Command: enable start of 3kV DC system		
2F_04.01.Zefiro-Europe.CONCEPT.905	Command: enable start of remote 15kV AC system		
2F_04.01.Zefiro-Europe.CONCEPT.906	Command: enable start of 15kV AC system		
2F_04.01.Zefiro-Europe.CONCEPT.907	Command: enable start of remote 25kV AC system		
2F_04.01.Zefiro-Europe.CONCEPT.908	Command: enable start of 25kV AC system		
2F_04.01.Zefiro-Europe.CONCEPT.910	Command: open line ciruit breaker		
2F_04.01.Zefiro-Europe.CONCEPT.911	Status: Line trip		
2F_04.01.Zefiro-Europe.CONCEPT.913	Command: isolate local AC system		
2F_04.01.Zefiro-Europe.CONCEPT.914	Max traction power due to transformer supervision		
2F_04.01.Zefiro-Europe.CONCEPT.915	Max line current RMS		
2F_04.01.Zefiro-Europe.CONCEPT.916	Command: AC line regeneration inhibit due to transformer supevision		
2F_04.01.Zefiro-Europe.CONCEPT.1106	Command: Isolate pantograph		2F_04.01.Zefiro-Europe.CONCEPT.531 2F_04.01.Zefiro-Europe.CONCEPT.762 2F_04.01.Zefiro-Europe.CONCEPT.763 2F_04.01.Zefiro-Europe.CONCEPT.995

TCMS SW to Supplier: CCU-O -> PCU

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.917	Country Code		
2F_04.01.Zefiro-Europe.CONCEPT.918	Status: neutral section		
2F_04.01.Zefiro-Europe.CONCEPT.919	Status: separation section		
2F_04.01.Zefiro-Europe.CONCEPT.934	Status: enable high voltage control from TCMS		

2F_04.01.Zefiro-Europe.CONCEPT.935	Status: Remote DC disconnector is closed		
2F_04.01.Zefiro-Europe.CONCEPT.936	Command: Enable start of 1500V DC from remote system		
2F_04.01.Zefiro-Europe.CONCEPT.937	Status: Remote DC high voltage system is isolated		
2F_04.01.Zefiro-Europe.CONCEPT.938	Command: Enable start of 1500V DC system from master PCU		
2F_04.01.Zefiro-Europe.CONCEPT.920	Status: Remote DC disconnector is closed		
2F_04.01.Zefiro-Europe.CONCEPT.921	Command: enable start of 3kV DC system from remote system		
2F_04.01.Zefiro-Europe.CONCEPT.922	Status: Remote DC high voltage system is isolated		
2F_04.01.Zefiro-Europe.CONCEPT.923	Command: Enable start of 3kV DC system from master PCU		
2F_04.01.Zefiro-Europe.CONCEPT.924	Status: Remote AC disconnector is closed		
2F_04.01.Zefiro-Europe.CONCEPT.925	Command: Enable start of 15kV AC from remote system		
2F_04.01.Zefiro-Europe.CONCEPT.926	Command: Enable start of 15kV AC system from master PCU		
2F_04.01.Zefiro-Europe.CONCEPT.927	Status: Remote AC disconnector switch is closed		
2F_04.01.Zefiro-Europe.CONCEPT.928	Command: Enable start of 25kV AC from remote system		
2F_04.01.Zefiro-Europe.CONCEPT.929	Command: Enable start of 25kV AC system from master PCU		
2F_04.01.Zefiro-Europe.CONCEPT.909	Status:All pantographs down		

Status: Line circuit breaker closed

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.930	Max traction power due to transformer supervision from redundant PCU on same transformer		
2F_04.01.Zefiro-Europe.CONCEPT.931	Max line circuit current from redundant PCU on same transformer, RMS		
2F_04.01.Zefiro-Europe.CONCEPT.932	Valid bit:communication with redundant PCU on same transformer		
2F_04.01.Zefiro-Europe.CONCEPT.933	Command: AC line regeneration inhibit due to transformer supervision from redundant PCU on same transformer		

Supplier to TCMS SW: Panto -> CCU-O

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.942	Pantograph 1 position		2F_04.01.Zefiro-Europe.CONCEPT.284
2F_04.01.Zefiro-Europe.CONCEPT.943	Pantograph 2 position		2F_04.01.Zefiro-Europe.CONCEPT.284
2F_04.01.Zefiro-Europe.CONCEPT.944	Pantograph 1 unintentionally raised		
2F_04.01.Zefiro-Europe.CONCEPT.945	Pantograph 2 unintentionally raised		
2F_04.01.Zefiro-Europe.CONCEPT.947	Pantograph 1 unintentionally lowered		2F_04.01.Zefiro-Europe.CONCEPT.207 2F_04.01.Zefiro-Europe.CONCEPT.309
2F_04.01.Zefiro-Europe.CONCEPT.948	Pantograph 2 unintentionally lowered		2F_04.01.Zefiro-Europe.CONCEPT.207 2F_04.01.Zefiro-Europe.CONCEPT.309
2F_04.01.Zefiro-Europe.CONCEPT.949	Pantograph 1 does not raise on command		2F_04.01.Zefiro-Europe.CONCEPT.207 2F_04.01.Zefiro-Europe.CONCEPT.309
2F_04.01.Zefiro-Europe.CONCEPT.950	Pantograph 2 does not raise on command		2F_04.01.Zefiro-Europe.CONCEPT.207 2F_04.01.Zefiro-Europe.CONCEPT.309
2F_04.01.Zefiro-Europe.CONCEPT.951	Pantograph 1 does not lower on command		
2F_04.01.Zefiro-Europe.CONCEPT.952	Pantograph 2 does not lower on command		
2F_04.01.Zefiro-Europe.CONCEPT.953	Pantograph 1 rise time too long		
2F_04.01.Zefiro-Europe.CONCEPT.954	Pantograph 2 rise time too long		
2F_04.01.Zefiro-Europe.CONCEPT.955	Pantograph 1 ADD tripped		2F_04.01.Zefiro-Europe.CONCEPT.294 2F_04.01.Zefiro-Europe.CONCEPT.332
2F_04.01.Zefiro-Europe.CONCEPT.956	Pantograph 2 ADD tripped		2F_04.01.Zefiro-Europe.CONCEPT.294 2F_04.01.Zefiro-Europe.CONCEPT.332
2F_04.01.Zefiro-Europe.CONCEPT.957	Pantograph 1 wear detection		2F_04.01.Zefiro-Europe.CONCEPT.1062 2F_04.01.Zefiro-Europe.CONCEPT.1063
2F_04.01.Zefiro-Europe.CONCEPT.958	Pantograph 2 wear detection		2F_04.01.Zefiro-Europe.CONCEPT.1061 2F_04.01.Zefiro-Europe.CONCEPT.1062

TCMS SW to Supplier: CCU-O -> Panto

Nr	Functional name	Type	Linked to requirement
2F_04.01.Zefiro-Europe.CONCEPT.960	Train speed		2F_04.01.Zefiro-Europe.CONCEPT.97
2F_04.01.Zefiro-Europe.CONCEPT.961	Selected Voltage		2F_04.01.Zefiro-Europe.CONCEPT.97
2F_04.01.Zefiro-Europe.CONCEPT.962	Selected catenary		2F_04.01.Zefiro-Europe.CONCEPT.97
2F_04.01.Zefiro-Europe.CONCEPT.963	Pantograph position command (raise order)		2F_04.01.Zefiro-Europe.CONCEPT.283

2F_04.01.Zefiro-Europe.CONCEPT.1120	Wear detection blocked (ICD Weardetec)		2F_04.01.Zefiro-Europe.CONCEPT.1119

5 Revision Log

5.1 Revision log 3EST000225-7907_B

Revision:	**_B**		
Date:	**2013-04-19**		
Editor:	**P. August**		
Change ID DOORS	**Refers to requirement ID**	**Title**	**Change content**
2F_04.01.ISSUES.-.Zefiro-Europe.318		Pantograph: wear detection indication on IDU	To be checked if a pictogram is needed to show panto carbon strip status and remaining distance that can be run, otherwise remove requirement to display remaining distance of worn pantograph on IDU **Deleted requirements:** 2F_04.01.Zefiro-Europe.CONCEPT.1108 2F_04.01.Zefiro-Europe.CONCEPT.1109 2F_04.01.Zefiro-Europe.CONCEPT.1065 ISSUE n° 735 in 4S TCMS SW
2F_04.01.ISSUES.-.Zefiro-Europe.319	[2F_04.01.Zefiro-Europe.CONCEPT.284]:	Panto status updated in Picture Model	Pantograph status update i Picture Model. Requirement 2F_04.01.Zefiro-Europe.CONCEPT.284 are not including all status required by PM. Acc to PM shall following status be shown: – Raised – Lowered – Cut out – Faulty raised – Faulty lowered Also what is the priority for TCMS SW? With the old status the TCMS SW had the following priority: – Cut out – Faulty/Blocked – Up – Down SCR on TCMS SW already created, ZEFIT00001023 CG 20130404: Created Issue 757 on 4S TCMS SW
2F_04.01.ISSUES.-.Zefiro-Europe.320	[2F_04.01.Zefiro-Europe.CONCEPT.1157]: [2F_04.01.Zefiro-Europe.CONCEPT.1158]:	ADD detection: Failure in circuitry commanding LCB open and panto fast down train line	In case of failure of relays in ADD detection circuitry (they command LCB open and panto fast down train line), by DI TCMS is able to recognize this event and block the raising command for the pantograph. (the driver can select the other pantograph, but to avoid stopping on railway in case both pantographs are faulty:) In emergency operation mode (see CTC), TCMS allows to raise pantograph, even though before this failure was detected.
2F_04.01.ISSUES.-.Zefiro-Europe.321	[2F_04.01.Zefiro-Europe.CONCEPT.1103]:	Pantograph disconnector unwanted opening	pantograph disconnector opens without order means: "closed" feedback disappears

Revision:	**_B**		
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Change ID DOORS	**Refers to requirement ID**	**Title**	**Change content**
2F_04.01.ISSUES.-.Zefiro-Europe.323	[2F_04.01.Zefiro-Europe.CONCEPT.1163]:	Parking mode auto exit: line voltage control	In case of previous missing line voltage parking mode auto exit procedure shall be stopped (and train shall return in parking mode) if line voltage is returned. LCB shall be automatically closed after a defined amount of time from the line voltage return that is proportional to the train number (to avoid simultaneous closing of many train LCBs) SEE ALSO CTC ISSUE #90
2F_04.01.ISSUES.-.Zefiro-Europe.324	[2F_04.01.Zefiro-Europe.CONCEPT.1160]: [2F_04.01.Zefiro-Europe.CONCEPT.1161]: [2F_04.01.Zefiro-Europe.CONCEPT.1162]: [2F_04.01.Zefiro-Europe.CONCEPT.1164]:	Double LCB closing during cab change in parking mode	SEE MR 305 During the procedure of “change desk” in parking mode the train needs to automatically reconfigure the pantograph layout when the other desk is enabled. In this sequence it is not allowed to have any interruption of the auxiliary power supply and passengers comfort (eg HVAC). It’s required that both LCB are closed for few seconds (approx 2sec) with both panto up to ensure the continuity of the line voltage power supply. After this the previous LCB opens and the related panto is lowered To avoid a possible short circuit TCMS shall check the voltage coherence of the two panto and after that it allows both LCB to be closed. Function will only be implemented when DC line voltage system is selected.
2F_04.01.ISSUES.-.Zefiro-Europe.325	[2F_04.01.Zefiro-Europe.CONCEPT.1115]: [2F_04.01.Zefiro-Europe.CONCEPT.1159]: [2F_04.01.Zefiro-Europe.CONCEPT.612]:	Panto raising/ lowering command	Update SFRS according MR 492
2F_04.01.ISSUES.-.Zefiro-Europe.328	[2F_04.01.Zefiro-Europe.CONCEPT.1152]:	Line current limits DC	Different levels for DC 3kV and DC 1.5kV system to be in line also with CONCEPT.728

5.2 Revision log 3EST000225-7907_C

Revision:	**_C**		
Date:	**2018-05-29**		
Editor:	**G. Parisi**		
Change ID DOORS	**Refers to requirement ID**	**Title**	**Change content**
2F_04.01.ISSUES.-.Zefiro-Europe.332	[2F_04.01.Zefiro-Europe.CONCEPT.315]: [2F_04.01.Zefiro-Europe.CONCEPT.578]:	LCB opening	Agreed with PPC to change voltage levels for LCB opening from >31.0kV to >=31.1kV and from >29.0kV to >=29.1kV
2F_04.01.ISSUES.-.Zefiro-Europe.334	[2F_04.01.Zefiro-Europe.CONCEPT.527]:	Propulsion set-up for DC 1,5kV	line voltage range for which propulsion system shall be set up for 1.5kV DC changed to ≥0.8 kV and <1.975 kV (agreed with PPC)
2F_04.01.ISSUES.-.Zefiro-Europe.335	[2F_04.01.Zefiro-Europe.CONCEPT.563]: [2F_04.01.Zefiro-Europe.CONCEPT.565]:	Control of pantograph disconnector	the disconnector will be closed at first, when the feedback that the disconnector is closed is present, the panto will be raised up. To lower the panto let the logic as it is.
2F_04.01.ISSUES.-.Zefiro-Europe.336	[2F_04.01.Zefiro-Europe.CONCEPT.315]: [2F_04.01.Zefiro-Europe.CONCEPT.316]: [2F_04.01.Zefiro-Europe.CONCEPT.319]: [2F_04.01.Zefiro-Europe.CONCEPT.503]: [2F_04.01.Zefiro-Europe.CONCEPT.539]: [2F_04.01.Zefiro-Europe.CONCEPT.540]: [2F_04.01.Zefiro-Europe.CONCEPT.578]: [2F_04.01.Zefiro-Europe.CONCEPT.579]: [2F_04.01.Zefiro-Europe.CONCEPT.580]: [2F_04.01.Zefiro-Europe.CONCEPT.581]: [2F_04.01.Zefiro-Europe.TRS.328]:	DC Line voltage supervision: 5min protection threshold	DC Voltage threshold set in DC line voltage 5min protection (3.6kV) is too low for normal values present on Italian overhead lines. Value must be changed to avoid unwanted LCB openings. Oliver Di Domenico: 10 min in 25 kV AC as in tender doc; added reset also when LCB is open; removed 5 min in 15 kV AC and in DC line because it is not contractual **in BL2.2 deleted requirements:** 2F_04.01.Zefiro-Europe.CONCEPT.502; 2F_04.01.Zefiro-Europe.CONCEPT.504
2F_04.01.ISSUES.-.Zefiro-Europe.337	[2F_04.01.Zefiro-Europe.CONCEPT.782]:	Inconsistency between pantograph status and position of toggle switch on driver's desk	Remove this requirement in order to avoid double message on IDU in many situation (e.g. ADD intervention, emergency brake by mushroom pushbutton, etc.) related to lowering pantograph without command from driver's desk If req.782 is approved to remove, then other requirements could set an EVENT. See ISSUES: 342 347 348

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2F_04.01.ISSUES.-.Zefiro-Europe.338	[2F_04.01.Zefiro-Europe.CONCEPT.1050]: [2F_04.01.Zefiro-Europe.CONCEPT.1152]: [2F_04.01.Zefiro-Europe.CONCEPT.728]:	Limit current on IDU too high	New limits and steps to be defined FCCB statement: For single train (one panto+ one LCB closed) a) in 25 KVAC line voltage systems: 560 A b) in DC 3 kV line voltage system: 3000 A c) in DC 1.5 kV line voltage system: 3200 A in case of 15kV line voltage system: 700 A In case of double traction the limits are doubled. Every time the catenary change, the max value for that catenary must be setted. current by enabling a current limiter device on the IDU: a) in AC line voltage systems by steps of 20 A (in range of 20 to 560 A) b) in DC line voltage systems by steps of 100 A (in range of 100 to 3000 A)
2F_04.01.ISSUES.-.Zefiro-Europe.339	[2F_04.01.Zefiro-Europe.CONCEPT.374]:	LCB closure from driver's desk: master controller in 0	LCB can be closed from driver's desk only if the master controller is on zero position. The management of LCB from ATP remains as it is.
2F_04.01.ISSUES.-.Zefiro-Europe.340	[2F_04.01.Zefiro-Europe.CONCEPT.1159]: [2F_04.01.Zefiro-Europe.CONCEPT.1165]: [2F_04.01.Zefiro-Europe.CONCEPT.1166]:	PCU detects incongruence between HW and SW 'raising signal'	PCU (Panto) detects incongruence between HW and SW 'raising signal' (= SW rise order is missing OR HW rise order is missing) while the pantograph is raised. After 5 s (if that signal has not been reset): – LCB shall be ordered to open (TCMS via SW); – the normal lowering sequence shall be started (raising command has been removed and then pantograph disconnector shall be order to open) After 10 s (if the pantograph has not been brought in folded position): – TCMS shall energize "LCB emergency open" train line; – then after 500 ms TCMS shall energize "Panto Fast down " train line; – after additional 10 s the pantograph disconnector shall be ordered to open TCMS sends to every PCU that "Panto Fast down " train line is energized (it means that EV3 inside PPU has been energized)
2F_04.01.ISSUES.-.Zefiro-Europe.341	[2F_04.01.Zefiro-Europe.CONCEPT.1167]:	ADD tripped signal from PCU	When the signal from PCU: "ADD tripped" is received: – TCMS shall energize "LCB emergency open" train line; – TCMS shall energize "Panto Fast down " train line and remove all raising commands. – after 10 s the pantograph disconnector shall be ordered to open (*this point is already covered by req. 565*)

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Change ID DOORS	**Refers to requirement ID**	**Title**	**Change content**
2F_04.01.ISSUES.-.Zefiro-Europe.342	[2F_04.01.Zefiro-Europe.CONCEPT.1097]: [2F_04.01.Zefiro-Europe.CONCEPT.1121]: [2F_04.01.Zefiro-Europe.CONCEPT.207]:	Pantograph is not raised on command	Pantograph is not raised when it is commanded to raise In normal conditions (i.e. not in back up mode, when PPU is directly driven from TCMS MIO DO), if the pantograph is not being declared from PCU as raised or moving within 15 s: – the raising command shall be removed; – it will be allowed to the driver to try for 3 times to raise again that pantograph, after that it shall be blocked and assumed as faulty (req.782 could be removed, see also ISSUES 337) 2014/05/26: after peer review, removed the limit on number of attempts
2F_04.01.ISSUES.-.Zefiro-Europe.343	[2F_04.01.Zefiro-Europe.CONCEPT.1153]:	Pantograph reported as in folded position without any command to lower	While pantograph is assumed to be raised, in case the signal from PCU states that this pantograph is not in raised position, but: – none has commanded to lower it AND – the PCU is not excluded (major fault or no life signal) AND – the pantograph is not raised in back up mode (i.e. PPU is not directly driven from TCMS MIO DO), then LCB has to be ordered to open by train line "LCB emergency OFF"
2F_04.01.ISSUES.-.Zefiro-Europe.344	[2F_04.01.Zefiro-Europe.CONCEPT.1157]: [2F_04.01.Zefiro-Europe.CONCEPT.1176]:	ADD trip due to EV3 commanded	ADD trip due to EV3 commanded To avoid false diagnosis: 1) PCU has to acquire the status of EV3 and consequently to mask the diagnostic signal "ADD tripped" if EV3 has been energized; 2) TCMS has to mask the diagnose and blocking condition: "ADD detection faulty" (by filtering it for >1 s if the "Panto Fast down" train line has been energized)
2F_04.01.ISSUES.-.Zefiro-Europe.345	[2F_04.01.Zefiro-Europe.CONCEPT.1168]:	Major Fault from PCU	If Major Fault from PCU has been detected in TCMS while the corresponding pantograph is raised, then "LCB emergency off" train line shall be immediately energized
2F_04.01.ISSUES.-.Zefiro-Europe.346	[2F_04.01.Zefiro-Europe.CONCEPT.1169]:	ADD detection by relay	In case ADD has been detected when TCMS reads the relay KPD, after 1 s the raising command from TCMS has to be removed

Revision:	**_C**		
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2F_04.01.ISSUES.-.Zefiro-Europe.347	[2F_04.01.Zefiro-Europe.CONCEPT.1067]: [2F_04.01.Zefiro-Europe.CONCEPT.1110]: [2F_04.01.Zefiro-Europe.CONCEPT.1170]: [2F_04.01.Zefiro-Europe.CONCEPT.1171]: [2F_04.01.Zefiro-Europe.CONCEPT.372]:	Reset of signal: ADD tripped from PCU	PCU resets the signal: ADD tripped after 20 s that the pantograph has been folded. Tuning in OTC: timing of relay KPD >= rather than PCU (e.g. 20 s) TCMS resets the signal: ADD tripped as soon as both signals (PCU and relay KPD) has gone to 0 TCMS allows 1 attempt in order to try to raise again this pantograph (at standstill ONLY) before to define it as blocked The number of attempts must be a parameter set to 0 at default value 2014-06-09/AS Signal "ADD tripped" from the pantograph control unit is no longer used in 1067, check if it shall be included in 1067 or 1169" 2014-06-09/GG: concept 1067 never stated to lower the pantograph due to ADD trip signal from PCU. It's a 4S issue. 2015/01/15 GP: the requirement has been changed as the predisposition of raising attempt will not be operational any time soon.
2F_04.01.ISSUES.-.Zefiro-Europe.348	[2F_04.01.Zefiro-Europe.CONCEPT.1172]: [2F_04.01.Zefiro-Europe.CONCEPT.1173]:	Missing IP communication between PCU and TCMS	In case the communication between PCU and TCMS has been missed, PCU is not allowed to block and make the pantograph lower. If TCMS doesn't see the life signal coming from PCU, after time out of 3-5 s it starts the following sequence: – LCB emergency open trainline shall be energized; – after 500 ms, Panto Fast down trainline shall be energized; – after 10 s the pantograph disconnector shall be ordered to open. TCMS declares this PCU as faulty (as in case of Major Fault) until there isn't the corresponding life signal
2F_04.01.ISSUES.-.Zefiro-Europe.349	[2F_04.01.Zefiro-Europe.CONCEPT.1174]: [2F_04.01.Zefiro-Europe.CONCEPT.1175]: [2F_04.01.Zefiro-Europe.CONCEPT.1185]:	Diagnosis of the contact: Major Fault (PCU)	Diagnosis of the contact: Major Fault (PCU) During the start up phase, after self test of PCU, when the life signal from PCU is being received, this test begins and TCMS shall manage it: – TCMS reads the contact: Major Failure in closed position (i.e no failure inside PCU); – TCMS sends to PCU the command to open (just for test) the contact: Major Failure; – TCMS reads the the contact: Major Failure in open position (i.e. it is not stuck); – TCMS sends to PCU the information that this test has been completed and the result.

Revision:	**_C**		
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2F_04.01.ISSUES.-.Zefiro-Europe.351	[2F_04.01.Zefiro-Europe.CONCEPT.1215]: [2F_04.01.Zefiro-Europe.CONCEPT.543]:	first raising of pantograph by auxiliary compressor	first raising of pantograph by auxiliary compressor: the time has to be increased time to fill auxiliary reservoire 0→6 bar in 127 s 0→4 bar in 75 s (without taking in account pipe volume) then 0→4 bar could be 80/85 sec taking in account pipe volume 12/01/2015 in order to avoid false faults at the pantograph first rising the requirement has been temporary changed: *If the air pressure to the pantograph is low (pressure switch) AND the main recevoir pressure is <6 bar; the order to raise the pantograph shall be set when:* – *pressure switch toggles to high* *OR* – *independently from the pressure switch after 6 minutes that the auxiliary compressor has been running continuously.*
2F_04.01.ISSUES.-.Zefiro-Europe.352	[2F_04.01.Zefiro-Europe.CONCEPT.1105]: [2F_04.01.Zefiro-Europe.CONCEPT.464]:	Losing of 3 kV with panto up	**Attenzione: il testo del MR è stato modificato il 12/05 → attendere approvazione MR per aggiornare SFRS con testo finale!** When the voltage supply is 3 kV and for some reasons there is no voltage applied, the propulsion system should open the main circuit breaker and open the System Isolation Switch. But the propulsion system has not to order to lower the panto. REA has to modify the SFRS.
2F_04.01.ISSUES.-.Zefiro-Europe.353	[2F_04.01.Zefiro-Europe.CONCEPT.1178]: [2F_04.01.Zefiro-Europe.CONCEPT.1179]: [2F_04.01.Zefiro-Europe.CONCEPT.1180]: [2F_04.01.Zefiro-Europe.CONCEPT.1181]: [2F_04.01.Zefiro-Europe.CONCEPT.1182]: [2F_04.01.Zefiro-Europe.CONCEPT.1183]:	LCB Em Off activation test	Trainline test is missing. Functionality has to be introduced. Test of LCB em off trainline can be done: – automatically every time after the LIM test is performed – when IDU soft key is pressed from maintainer
2F_04.01.ISSUES.-.Zefiro-Europe.354	[2F_04.01.Zefiro-Europe.CONCEPT.1177]:	Maintenance mode relays diagnosis	In order to fulfill the req. 495, a maintenance mode relay in each DM car is needed. It is enough that just one F key multiplier (either on DM1 car or on DM8 car) to have maintenance mode relay energized both on DM1 car and DM8 car. A diagnostic function to monitor these maintenance mode relays will be added: "An event shall be set in case the maintenance mode relays have discrepancy between DM1 and DM8 on the same trainset"

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2F_04.01.ISSUES.-.Zefiro-Europe.355	[2F_04.01.Zefiro-Europe.CONCEPT.1124]: [2F_04.01.Zefiro-Europe.CONCEPT.1125]: [2F_04.01.Zefiro-Europe.CONCEPT.344]: [2F_04.01.Zefiro-Europe.CONCEPT.443]: [2F_04.01.Zefiro-Europe.CONCEPT.776]:	LCB manually opened while passing on neutral section	If LCB is already open when ATP set opening command (e.g before a neutral section), Automatic re-closure shall be avoided
2F_04.01.ISSUES.-.Zefiro-Europe.356	[2F_04.01.Zefiro-Europe.CONCEPT.1189]: [2F_04.01.Zefiro-Europe.CONCEPT.1190]: [2F_04.01.Zefiro-Europe.CONCEPT.1191]: [2F_04.01.Zefiro-Europe.CONCEPT.1192]: [2F_04.01.Zefiro-Europe.CONCEPT.1193]: [2F_04.01.Zefiro-Europe.CONCEPT.1194]: [2F_04.01.Zefiro-Europe.CONCEPT.1195]: [2F_04.01.Zefiro-Europe.CONCEPT.1196]: [2F_04.01.Zefiro-Europe.CONCEPT.1197]: [2F_04.01.Zefiro-Europe.CONCEPT.1198]: [2F_04.01.Zefiro-Europe.CONCEPT.1199]: [2F_04.01.Zefiro-Europe.CONCEPT.1200]: [2F_04.01.Zefiro-Europe.CONCEPT.1201]: [2F_04.01.Zefiro-Europe.CONCEPT.1202]: [2F_04.01.Zefiro-Europe.CONCEPT.1203]: [2F_04.01.Zefiro-Europe.CONCEPT.1204]: [2F_04.01.Zefiro-Europe.CONCEPT.1205]: [2F_04.01.Zefiro-Europe.CONCEPT.1206]: [2F_04.01.Zefiro-Europe.CONCEPT.1207]: [2F_04.01.Zefiro-Europe.CONCEPT.1208]: [2F_04.01.Zefiro-Europe.CONCEPT.1209]: [2F_04.01.Zefiro-Europe.CONCEPT.1210]: [2F_04.01.Zefiro-Europe.CONCEPT.1211]: [2F_04.01.Zefiro-Europe.CONCEPT.1212]:	HVAC pre-conditioning on DC line	HVAC pre-conditioning on DC line

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	[2F_04.01.Zefiro-Europe.CONCEPT.1213]:		
2F_04.01.ISSUES.-.Zefiro-Europe.357	[2F_04.01.Zefiro-Europe.CONCEPT.310]:		In order to avoid false diagnosis on the pantograph rising/lowering phase the timing changes from 10s to 20 sec. Moreover, due to the fact that raised and lowered switch are not foreseen from the system also the logic to set the event is uptodated.
2F_04.01.ISSUES.-.Zefiro-Europe.358	[2F_04.01.Zefiro-Europe.CONCEPT.782]:		The event is only useful at the first cab activation, as any other pantograph lowering condition is already monitored.
2F_04.01.ISSUES.-.Zefiro-Europe.359	[2F_04.01.Zefiro-Europe.CONCEPT.1214]:		**FROM AUXILIARY ISSUE: 2F_06.01.ISSUES.-.Zefiro-Europe.140:** Traction block will be removed since is not useful (plug inserted cannot be detected by mechanical switch) Requirement to be updated in following way: – if any plug is detected inserted with train running OR HV already enabled, affected ext.plug shall be constantly disabled by TCMS (till Master reset) – in case (with train standtill) driver try to raise panto, a warning shall be set advicing thatn plug is reported inserted (IDU guide shall state that in case of false reporting, plug shall be manually isolated using softkey on iDU) – an isolation command of each ext plug shall be added on IDU. when this command is set by user, TCMS shall constantly energize DO to isolate that plug. ***MOREOVER the following req has been added: 2F_04.01.Zefiro-Europe.CONCEPT.1214***
2F_04.01.ISSUES.-.Zefiro-Europe.360	[2F_04.01.Zefiro-Europe.CONCEPT.1121]:		Due to too short timeout the pantograph goes faulty without any reason. Parameter to be increased from 5 minutes to 10 minutes
2F_04.01.ISSUES.-.Zefiro-Europe.361	[2F_04.01.Zefiro-Europe.CONCEPT.421]:		in order to solve ID99 reporting of a contradictory logic. The System Isolation SW (4a) has to be open and then closed for a successful earthing procedure.
2F_04.01.ISSUES.-.Zefiro-Europe.362	[2F_04.01.Zefiro-Europe.CONCEPT.492]:		PPC reads autonomously the line voltage via the system detection - ID55 open points TS5
2F_04.01.ISSUES.-.Zefiro-Europe.363	[2F_04.01.Zefiro-Europe.CONCEPT.1216]: [2F_04.01.Zefiro-Europe.CONCEPT.1217]:		due to automatic LCB closing request the TCMS has to supervise the line voltage level and request the LCB closure only if it is within the right levels
2F_04.01.ISSUES.-.Zefiro-Europe.364	[2F_04.01.Zefiro-Europe.CONCEPT.1218]:		new req.1218. when A-key is restored the MV contactors remain open (unless the train is switched off and then on again). For this reason when the A-key is restored to 'normal mode' position (from 'Depot mode') all MV contactors 8, 18c and 39 shall be closed.
2F_04.01.ISSUES.-.Zefiro-Europe.365	[2F_04.01.Zefiro-Europe.CONCEPT.1172]:		the requirement shall be based on the communication error timing rather than requesting the implementation of new logics with new and different timings

Revision:	**_C**		
Date:	**2018-05-29**		
Editor:	**G. Parisi**		
Change ID DOORS	**Refers to requirement ID**	**Title**	**Change content**
2F_04.01.ISSUES.-.Zefiro-Europe.366	[2F_04.01.Zefiro-Europe.CONCEPT.1097]: [2F_04.01.Zefiro-Europe.CONCEPT.1121]: [2F_04.01.Zefiro-Europe.CONCEPT.207]:		– req. 1121 shall be done within the air supply module to declare the 110V aux compressor faulty → DELETED – req 1097 is already implemented within the PCU – req 207 is useless as there is no requirement (except for ADD trip) blocking the panto
2F_04.01.ISSUES.-.Zefiro-Europe.368	[2F_04.01.Zefiro-Europe.CONCEPT.559]:		Ice scraping NOT allowed in case of pantograph command in backup mode
2F_04.01.ISSUES.-.Zefiro-Europe.369	[2F_04.01.Zefiro-Europe.CONCEPT.1219]: [2F_04.01.Zefiro-Europe.CONCEPT.1220]:		in order to filter false diagnosis of 'panto unknown' and avoid fake panto fault diagnosis, an inhibition criteria is introduced between a lowering order and a rising order of the same panto
2F_04.01.ISSUES.-.Zefiro-Europe.370	[2F_04.01.Zefiro-Europe.CONCEPT.1124]: [2F_04.01.Zefiro-Europe.CONCEPT.1125]:		removed concepts 1124 and 1125, they are already covered by the other concepts 344, 776 and 443. (SFRS concept 1124, 1125 and 443 are all linked only to 4s TRS TCMS 284, so it is enough to keep concept 443)
2F_04.01.ISSUES.-.Zefiro-Europe.371	[2F_04.01.Zefiro-Europe.CONCEPT.1169]:		clarification of requirement: panto UP order should be removed as soon as the ADD trip is detected
2F_04.01.ISSUES.-.Zefiro-Europe.372	[2F_04.01.Zefiro-Europe.CONCEPT.1000]: [2F_04.01.Zefiro-Europe.CONCEPT.1222]:		in DC mode: the pantograph has to be blocked in case the System isolation switch goes faulty and not in case the roof disconnectors faulty in AC mode: the pantograph has to be blocked in case roof disconnectors faulty
2F_04.01.ISSUES.-.Zefiro-Europe.373	[2F_04.01.Zefiro-Europe.CONCEPT.1091]: [2F_04.01.Zefiro-Europe.CONCEPT.1093]:		In order to solve the scenario for which with one roof disconnector plausibility fault the LCB could not be closed the management of the DC roof disconnector has been changed.
2F_04.01.ISSUES.-.Zefiro-Europe.374	[2F_04.01.Zefiro-Europe.CONCEPT.1224]:		when LCB is requested (from driver) to close while the Master controller is out of the 0 position, an advisory shall be shown to the driver asking 'to set the master controller in 0 position before ordering LCB closure'
2F_04.01.ISSUES.-.Zefiro-Europe.376	[2F_04.01.Zefiro-Europe.CONCEPT.1153]:		In case pantograph is lowered without order (signal from PCU) the LCB shall be opened by activation of "LCB emergency open" trainline, the pantograph order shall be removed and an alarm shall be set according to the diagnosis from the PCU.
2F_04.01.ISSUES.-.Zefiro-Europe.377	[2F_04.01.Zefiro-Europe.CONCEPT.309]:		In case the pantograph does not lower on command the diagnosis of the PCU shall rise an alarm to the driver and the pantograph shall be blocked.
2F_04.01.ISSUES.-.Zefiro-Europe.378	[2F_04.01.Zefiro-Europe.CONCEPT.1220]: [2F_04.01.Zefiro-Europe.CONCEPT.1225]:		text of req.1220 has been changed (independent from time) new req has been added to tell the driver that the pantograph is re-risable

Revision:	**_C**		
Date:	**2018-05-29**		
Editor:	**G. Parisi**		
Change ID DOORS	**Refers to requirement ID**	**Title**	**Change content**
2F_04.01.ISSUES.-.Zefiro-Europe.379	[2F_04.01.Zefiro-Europe.CONCEPT.1165]:		In case Pantograph Control unit (PCU) detects incongruence between HW and SW 'raising signal' (= SW rise order is missing OR HW rise order is missing) while the pantograph is raised, the following actions shall be carried out. After 2 s from incoherence detection: – LCB shall be ordered to open (TCMS via SW); – the pantograph shall be lowered and blocked – an event shall be set to the driver (Change pantograph required) – the related fault event shall be set to the maintainer (incoherence between HW and SW orders or vice-versa) After 16 s from incoherence detection (if the pantograph has not been lowered): – TCMS shall energize "Panto Fast down " train line;
2F_04.01.ISSUES.-.Zefiro-Europe.380	[2F_04.01.Zefiro-Europe.CONCEPT.1226]: [2F_04.01.Zefiro-Europe.CONCEPT.1227]: [2F_04.01.Zefiro-Europe.CONCEPT.1228]: [2F_04.01.Zefiro-Europe.CONCEPT.1229]: [2F_04.01.Zefiro-Europe.CONCEPT.1230]: [2F_04.01.Zefiro-Europe.CONCEPT.1231]: [2F_04.01.Zefiro-Europe.CONCEPT.1232]: [2F_04.01.Zefiro-Europe.CONCEPT.1233]: [2F_04.01.Zefiro-Europe.CONCEPT.1234]:		New requirements to let the driver a manual opening of the roof disconnectors
2F_04.01.ISSUES.-.Zefiro-Europe.382	[2F_04.01.Zefiro-Europe.CONCEPT.989]:		Lim self test at the parking exit
2F_04.01.ISSUES.-.Zefiro-Europe.383	[2F_04.01.Zefiro-Europe.CONCEPT.1236]: [2F_04.01.Zefiro-Europe.CONCEPT.1237]:		Major fault test to be performed also in case of master reset An event shall be raised when major failure test is active
2F_04.01.ISSUES.-.Zefiro-Europe.384	[2F_04.01.Zefiro-Europe.CONCEPT.1113]:		When the backup DO order is energized the EV1 (EV2) order shall be notified to the PCU via IP
2F_04.01.ISSUES.-.Zefiro-Europe.385	[2F_04.01.Zefiro-Europe.CONCEPT.1165]: [2F_04.01.Zefiro-Europe.CONCEPT.1169]: [2F_04.01.Zefiro-Europe.CONCEPT.1171]: [2F_04.01.Zefiro-Europe.CONCEPT.1185]:		1. the major fault test will not lead to failure pf the pantograph in case of failure of the test 2. ADD trip will be only based on pressure switch PS3 and not anymore on PS2 - requirement 1167 deleted 3. In case of missing SW order the pantograph will not be ordered to lower
2F_04.01.ISSUES.-.Zefiro-Europe.386	[2F_04.01.Zefiro-Europe.CONCEPT.986]:		The LCB status lamp is not working correctly in Multiple unit see point 290 FCCB
2F_04.01.ISSUES.-.Zefiro-Europe.390	[2F_04.01.Zefiro-Europe.CONCEPT.1151]:	MR 1703	LIM by pass in multiple composition
2F_04.01.ISSUES.-.Zefiro-Europe.391	[2F_04.01.Zefiro-Europe.CONCEPT.548]:		speed limit removed from ice scraping mode due to test performed on line - even will be only 'ice scraping active'

Revision:	**_C**		
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Change ID DOORS	**Refers to requirement ID**	**Title**	**Change content**
2F_04.01.ISSUES.-.Zefiro-Europe.400	[2F_04.01.Zefiro-Europe.CONCEPT.1226]:		New requirements to let the driver a manual opening of the roof disconnectors
2F_04.01.ISSUES.-.Zefiro-Europe.401	[2F_04.01.Zefiro-Europe.CONCEPT.1238]:		due to Trenitalia request a new message for the driver shall be implemented to warn the driver to activate ice scraping mode
2F_04.01.ISSUES.-.Zefiro-Europe.403	[2F_04.01.Zefiro-Europe.CONCEPT.1245]:		AC LCB opening supervision